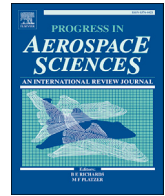




Contents lists available at ScienceDirect

Progress in Aerospace Sciences

journal homepage: www.elsevier.com/locate/paerosci

A review of cooperative and uncooperative spacecraft pose determination techniques for close-proximity operations

Roberto Opromolla^{*}, Giancarmine Fasano, Giancarlo Rufino, Michele Grassi

University of Naples "Federico II", Dept. of Industrial Engineering, P.le Tecchio 80, 80125, Napoli, Italy

ARTICLE INFO

Keywords:

Spacecraft pose determination
Close-proximity flight
Relative navigation
Electro-optical sensors
Cooperative targets
Uncooperative known targets
Uncooperative unknown targets

ABSTRACT

The capability of an active spacecraft to accurately estimate its relative position and attitude (pose) with respect to an active/inactive, artificial/natural space object (target) orbiting in close-proximity is required to carry out various activities like formation flying, on-orbit servicing, active debris removal, and space exploration. According to the specific mission scenario, the pose determination task involves both theoretical and technological challenges related to the search for the most suitable algorithmic solution and sensor architecture, respectively. As regards the latter aspect, electro-optical sensors represent the best option as their use is compatible with mass and power limitation of micro and small satellites, and their measurements can be processed to estimate all the pose parameters. Overall, the degree of complexity of the challenges related to pose determination largely varies depending on the nature of the targets, which may be actively/passively cooperative, uncooperative but known, or uncooperative and unknown space objects. In this respect, while cooperative pose determination has been successfully demonstrated in orbit, the uncooperative case is still under study by universities, research centers, space agencies and private companies. However, in both the cases, the demand for space applications involving relative navigation maneuvers, also in close-proximity, for which pose determination capabilities are mandatory, is significantly increasing. In this framework, a review of state-of-the-art techniques and algorithms developed in the last decades for cooperative and uncooperative pose determination by processing data provided by electro-optical sensors is herein presented. Specifically, their main advantages and drawbacks in terms of achieved performance, computational complexity, and sensitivity to variability of pose and target geometry, are highlighted.

1. Introduction

Spacecraft pose determination is the problem of computing the relative position and attitude of an active satellite (chaser) with respect to a target space object. Specifically, the target pose is unambiguously identified by a set of six parameters representing the translational and rotational Degrees of Freedom (DOF) of the relative motion. The estimated pose can be used as aiding information within a filtering scheme performing relative navigation, i.e., determining the relative state of the chaser with respect to the target. Based on this definition, spacecraft pose determination is clearly a relevant problem for a wide range of mission scenarios, like formation flying (FF) [1,2], on-orbit servicing (OOS) [3,4], active debris removal (ADR) [5,6], comet and asteroid exploration [7,8], which involve relative navigation maneuvers to be performed autonomously, i.e., without control from ground. Of course, the interest in pose determination is limited to close-proximity operations, such as close-

range rendezvous, final approach and docking, station-keeping, monitoring, or other coordinated maneuvers, which typically involve relative distances ranging from a few meters up to several tens of meters (depending on the target size) [9]. Indeed, at larger distances (e.g., during far- and mid-range rendezvous maneuvers) pose determination is not possible, and relative navigation can be performed using range and line-of-sight (LOS) information [10], or only LOS measurements [11].

When operating in close-proximity, the need for autonomy is motivated by the fact that communication delays or lack of coverage make relative navigation maneuvers based on ground control station commands unfeasible. This is the main reason why pose determination is a complex task requiring ad-hoc algorithmic and technological solutions. Specifically, some of the main factors driving the choice of pose determination algorithms and sensors are the mission scenario and the type of target in terms of cooperativeness.

A target space object is deemed cooperative if it is built to provide

^{*} Corresponding author.

E-mail addresses: roberto.opromolla@unina.it (R. Opromolla), giancarmine.fasano@unina.it (G. Fasano), giancarlo.rufino@unina.it (G. Rufino), michele.grassi@unina.it (M. Grassi).

<http://dx.doi.org/10.1016/j.paerosci.2017.07.001>

Received 28 February 2017; Received in revised form 29 June 2017; Accepted 18 July 2017

Available online xxxx

0376-0421/© 2017 Elsevier Ltd. All rights reserved.

information suitable for the estimation of its pose with respect to the chaser. Also, it can be actively or passively cooperative depending on whether it interacts with a dedicated radio-link with the chaser, or not. Cooperative targets are typical of both FF and OOS applications. However, a difference can be highlighted, since OOS always involves a servicing spacecraft as chaser and a serviceable platform as target [10], while in FF scenarios, besides this standard configuration [12,13], all the components of the formation may be purposely designed to cooperate with the others in terms of relative navigation [14], thus taking both the chaser and target roles.

Uncooperative targets, instead, can be divided in two categories, depending on whether at least geometrical information about their shape and size is available, or they are fully unknown. The former case can occur for OOS and ADR if the target (i.e., an operating satellite or a debris, respectively) is already on orbit when the servicing/removal mission is conceived, provided that at least a simplified model of the target is available. However, some ADR missions as well as comet and asteroid exploration scenarios require dealing with partially or fully unknown targets.

As mentioned above, the target cooperativeness has an impact on the selection of the specific sensor architecture. Cooperative pose determination may be carried out by exploiting sets of Radio-Frequency (RF) antennas installed on board of the chaser and the target. This concept was largely exploited at the beginning of the rendezvous era driven by the American and Russian space programs [15]. The example given by the Russian Kurs system, which was used for the rendezvous of the Soyuz and Progress vehicles toward the Russian space station Mir, first, and then to the International Space Station (ISS), is particularly relevant [16]. Indeed, different RF-based measurement principles can be exploited to obtain full pose information about the target, i.e., range, range-rate, LOS angles, and relative attitude parameters, while operating from a relative distance of several kilometers down to contact [9]. These RF technologies are nowadays outdated for 6-DOF pose determination, especially considering space applications based on small satellites, due to the strict requirements regarding measurement performance, mass and power budgets. Indeed, the Formation Flying Radio Frequency metrology subsystem recently tested on-orbit during the PRISMA mission is designed to provide only range and LOS measurements at 1 Hz with a typical accuracy of 1 cm and 1° [12]. Another solution in which the target is considered actively cooperative consists in the use of satellite navigation technology. Specifically, it requires the presence of Global Navigation Satellite System (GNSS) receivers and antennas on board of the chaser and the target, as well as a reliable communication link between the two satellites to exchange measurements. This technology can be used to obtain relative position measurements by exploiting the Differential GPS (DGPS) or the Carrier-phase Differential GPS (CDGPS) approaches [17], as demonstrated in space, for instance, during the PRISMA mission [18]. However, despite the high accuracy level attainable (up to meter- and centimeter-level for the DGPS and CDGPS approaches, respectively), GNSS-based relative navigation does not provide any information about the relative attitude, so the 6-DOF pose cannot be determined. In addition, the relative position measurements may be affected by larger errors due to multipath phenomena and partial occlusion of GNSS signals when operating in close-proximity. Hence, GNSS is the most valuable option for the relative positioning of cooperative targets but in mid/far range rendezvous scenarios. Given the limitations of RF-based and GNSS-based technologies, Electro-Optical (EO) sensors represent the best option to ensure pose determination in close-proximity. Indeed, these devices, which can be active, e.g., Light Detection and Ranging (LIDAR) systems, or passive, e.g., monocular and stereo cameras, are able to estimate the 6-DOF pose of a chaser orbiting in close-proximity to a space target provided that the overall target shape (or just a specific portion of it) is imaged in the sensor Field-of-View (FOV) with enough details [19]. This condition has impact on the specific range of relative distances where each type of EO sensor is better suited to work [6]. In view of the use of EO sensors for cooperative pose determination, a target space object can

be made passively cooperative if it is equipped with a set of artificial markers which can be easily detected and recognized in the acquired datasets. These markers can be active or passive, e.g., Light Emission Diodes (LEDs) or corner-cube reflectors (CCR), respectively, and they are located on the target surfaces according to specific known patterns. The use of EO sensors has high relevance also because they represent the only valid technological option for pose determination and relative navigation in close-proximity to uncooperative targets. Of course, this latter task is more challenging not only due to the absence of cooperative information but especially in view of the complex rotational dynamics (free tumbling) typical of natural/man-made uncontrolled space objects. Indeed, this phenomenon produces an increase of the collision risk when the chaser is moving close to the target, e.g., for monitoring or approach purposes. Hence, the accuracy requirements regarding the estimated pose and the relative navigation solution become more stringent to adequately assist the control of the relative motion. Another important aspect, which comes out when dealing with uncooperative targets, is that debris or operating satellites to be serviced may have suffered physical damages as well as optical degradation of their surfaces due to the prolonged exposure to the space environment, thus appearing different than expected. This issue may be tackled by foreseeing specific strategies to update or build a target model in real-time on board.

A summary of the technological solutions suitable for spacecraft relative navigation and of the corresponding potential mission scenarios depending on the target cooperativeness is presented in Table 1, where the symbol N/A means “not applicable”.

In the presented framework, the aim of this paper is to provide a detailed overview of the state-of-the-art on techniques and algorithms for spacecraft pose determination. In particular, the attention is focused on algorithms able to process data from active/passive EO sensors since they allow estimating the 6-DOF pose parameters regardless of the target cooperativeness, thus being suitable for any of the mentioned mission scenarios. Also, their mass and power consumption are compatible with the installation on board of micro or small satellites. Of course, a fundamental distinction is made between cooperative and uncooperative approaches, which are both of high interest to the space community.

Cooperative pose determination was successfully demonstrated on orbit [10,12] but the interest of space agencies and private companies to launch and operate spacecraft formations or to inspect, maintain and repair satellites (which are already on orbit) to extend their operative life is significantly growing, especially in view of the related economic benefit [20,21]. At the same time, several strategies were proposed to deal with the issue of pose determination of uncooperative targets, and their performance was assessed by carrying out numerical analyses and/or experimental test campaigns. However, the growth of debris population in orbit is significantly pushing universities, research centers, space agencies and private companies to continue investing large resources to identify robust technical and technological solutions and demonstrate

Table 1
Taxonomy of spaceborne relative navigation approaches and scenarios.

Target type	Relative navigation approach	Relative navigation hardware (chaser)	Relative navigation hardware (target)	Possible mission scenario
Actively cooperative	RF-based	RF transmitting/receiving antennas	RF transmitting/receiving antennas	FF, OOS
	GNSS-based	GNSS receiver and communication link	GNSS receiver and communication link	
Passively cooperative	EO-based	Monocular/stereo camera, LIDAR	Artificial markers, e.g., LEDs, CCRs	FF, OOS
Uncooperative known	EO-based	Monocular/stereo camera, LIDAR	N/A	OOS, ADR
Uncooperative unknown	EO-based	Monocular/stereo camera, LIDAR	N/A	ADR, comet/asteroid exploration

them on orbit as soon as possible. Indeed, this is mandatory to secure the possibility of future space activities.

The paper is organized as follows. Section 2 presents the main technical specifications, advantages and drawbacks related to the different typologies of state-of-the-art EO sensors suitable for spaceborne pose determination. The subsequent sections contain a detailed review of pose determination algorithms which have been developed to deal with cooperative (Section 3) and uncooperative targets either known (Section 4) or not (Section 5). Different approaches from the open literature are described by highlighting the main methodological aspects, the achieved results and the open challenges. Finally, a conclusion is drawn in Section 6.

2. Electro-optical sensors for pose determination: a survey

Generally speaking, the term “EO sensors” is used to indicate a large variety of devices capable of collecting radiations reflected and/or directly emitted by the surrounding environment in the optical spectrum. This portion of the electromagnetic spectrum approximately ranges from $0.01\ \mu\text{m}$ to $1000\ \mu\text{m}$, thus EO sensors are mainly classified (according to the specific band in which their detector is sensitive) as ultraviolet (between $0.01\ \mu\text{m}$ and $0.37\ \mu\text{m}$), visible (between $0.37\ \mu\text{m}$ and $0.75\ \mu\text{m}$) and infrared (between $0.75\ \mu\text{m}$ and $1000\ \mu\text{m}$) [22]. Also, they can be active or passive depending on whether they include an energy source to emit radiation or not.

Focusing on spaceborne applications, while ultraviolet systems are mainly tailored to astronomic observation, spacecraft relative navigation relies on visible and infrared sensors which can be classified into passive systems, either based on single (monocular) or multiple (stereo) cameras, and active LIDAR. As regards the former category, monocular and stereo cameras operating in the visible band are able to acquire data which can be processed to carry out pose determination. Instead, infrared images, though allowing target LOS estimation at far range [23], are too poorly textured for pose determination purposes, when acquired in close-proximity. On the other hand, the term “LIDAR” indicates a large variety of sensors that are basically able to measure distances by illuminating a target with an infrared laser source and analyzing the backscattered radiation. Unlike laser range finders (which use a single detector to

measure the distance travelled by a non-steerable laser beam, thus being able to compute exclusively range and LOS of the target), LIDARs produce sets of 3D data (point clouds) which can be exploited for pose determination, by collecting multiple echoes coming from an assigned FOV.

This discussion about the potential of EO sensors for spacecraft relative navigation and pose determination is summarized in Fig. 1.

The design of accurate and robust algorithms for cooperative or uncooperative pose determination is strictly related to the technological solution selected for the sensor. A concise summary of the main advantages and drawbacks related to the selection of active or passive EO sensors to carry out relative navigation in close-proximity is provided below.

Passive EO sensors are characterized by lower hardware complexity, cost and power consumption with respect to LIDARs, and they can be simultaneously used for supervised applications, like ground control station situational awareness and human-in-the-loop operations. Despite these important advantages, the high sensitivity to the largely variable illumination conditions typical of the space environment, and to the potential presence of other celestial bodies in their field of view, makes the use of passive cameras as main relative navigation sensor extremely challenging. On the other hand, LIDARs are robust to operations in poor visibility conditions and allow the target to be easily discriminated from the background (segmentation). However, the limited operational applicability of passive monocular and stereo cameras in poorly illuminated scenarios can be partially compensated by including light sources within the sensor architecture (thus making the system active). Of course, this choice involves an increase in the amount of electrical power required on board (though the power budget is typically expected to remain lower with respect to the use of LIDARs). In terms of output, stereo cameras can acquire 3D information about the target (range images) just like LIDARs. Although this involves additional processing steps, e.g., image rectification and disparity map calculation, range images are typically much denser (and consequently more textured) than LIDAR point clouds. However, the maximum range at which stereo performance is acceptable in terms of depth resolution (which determines the achievable accuracy in 3D feature detection), is strongly limited by the baseline of the stereo configuration [24]. Instead, LIDARs can ensure

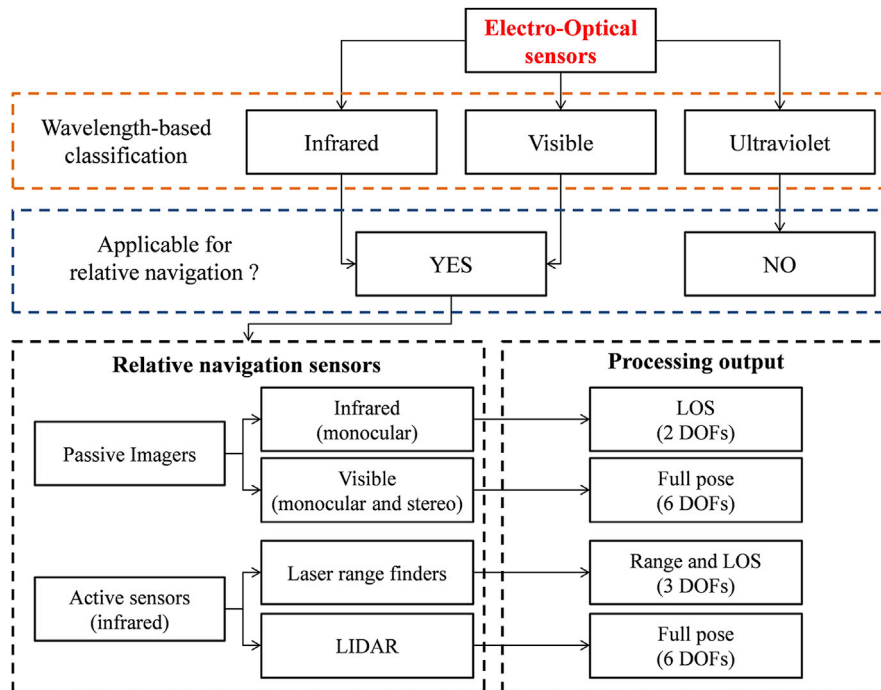


Fig. 1. Taxonomy of EO sensors for spacecraft applications.

very large operational ranges with constant accuracy level. Also, LIDARs can provide both 3D position and intensity data. On one side, this involves the necessity to deal with extremely large amounts of data, which can be too computationally burdensome (especially for applications on small satellites which typically have limited processing resources). On the other side, this richness in data can be exploited to compensate their typically poor spatial resolution.

In view of the above considerations, a trade-off between active and passive EO sensors should take several aspects into account, e.g., the mission scenario and the resources available on board of the chaser in terms of payload, electrical and processing power. For instance, a comprehensive analysis of the key features characterizing several capture and relative navigation technologies envisaged by Astrium for ADR can be found in Ref. [5]. The next sub-sections include detailed information about the main technical specifications of monocular and stereo cameras, on one side, and LIDARs, on the other side, tested on ground and/or operated on orbit to carry out cooperative and uncooperative pose determination.

2.1. Monocular sensors and stereo cameras

In the last two decades, three different generations of vision-based systems able to estimate the 6-DOF pose parameters of a cooperative target, were developed and tested in space to demonstrate OOS and FF capabilities. First, the Proximity Operation Sensor (PXS), designed by the National Space Development Agency of Japan (NASDA), was used for relative navigation during the docking phase in the framework of the 7th mission of the Engineering Test Satellite Program (ETS-VII) launched in 1999 [25–27]. Specifically, this system was able to estimate the relative position and attitude between the chaser's and target's docking frames. The PXS consists of a single visible camera on the chaser, and a set of seven non-coplanar, passive, round shaped markers, installed on the docking interface of the target. In order to cope with limited performance of the sensor when operating in poor visibility conditions, an array of LEDs, emitting pulsed visible light (640-nm wavelength) within a cone of 30°, is installed on the chaser to illuminate the docking interface. Detailed information about the sensor configuration and the achieved mission results can be found in Ref. [27]. Second, the Advanced Video Guidance Sensor (AVGS), developed by the Marshall Space Flight Center (MSFC), was one of the main relative navigation sensors during the Demonstration of Autonomous Rendezvous and Docking (DART) mission in 2005 [28], as well as the primary docking sensor during the Orbital Express (OE) mission in 2007 [29]. While DART achieved only partial results due to a failure of the navigation system which caused a collision between the chaser and the target before the nominal end of the mission [30], OE was completely successful. In particular, the AVGS was able to satisfy all the pre-defined accuracy requirement [31]. The sensor is composed of a visible camera on the chaser and two sets of four non-coplanar CCRs on the target (long-range and short range CCRs). Moreover, the AVGS includes two sets of LEDs (operating with 808-nm and 845-nm wavelength, respectively) to illuminate the CCRs. This solution, which had been similarly adopted for the PXS, allows obtaining high-intensity pixels on the image plane from the CCRs, even in poor visibility conditions. Details about the sensor configuration can be found in Ref. [32]. Finally, the Visual Based System (VBS), developed by the Technical University of Denmark (DTU), was one of the relative navigation technologies for FF applications tested during the PRISMA mission [33]. It consists of two cameras on the chaser and a set of active markers (LEDs) on the target. Specifically, five LEDs are located on each of the target's panels according to a coplanar, non-collinear pattern. While a far-range camera is used for LOS-only navigation from approximately 100 km down to 100 m of relative distance [34], a short-range camera is exploited to carry out pose determination, by imaging the above-mentioned LEDs, in the interval of distances that goes from a few tens of meters to few centimeters [35]. In order to enhance the LEDs visibility, a band pass filter centered on the wavelength of the emitted radiation is

part of the optical system of the short-range camera. A detailed analysis of the pose determination results obtained by processing the VBS images can be found in Refs. [36,37].

The main specifications regarding the above-mentioned systems are collected in Table 2, where the symbol *N/F* ("not found") indicates information not documented in the literature (to the authors' knowledge). Unlike PXS, which covers only the final approach and docking maneuvers, AVGS and VBS are characterized by a much larger operational range thus being suitable for close-range rendezvous. VBS requires less mass and electric power than AVGS and PXS, since no illumination sources are installed on the chaser. Of course, the limited electrical power required by the active markers contributes to the power-budget in the design of the cooperative target.

This survey shows that monocular-based sensors operating in the visible band are suitable for pose determination of cooperative targets in space while ensuring wide-enough operational ranges. As regards the possibility of using stereovision systems, instead, results of a feasibility study funded by the Italian Space Agency for a FF/OOS demonstration mission involving two small cooperative satellites (below 1 m and 100 kg for size and mass, respectively) are presented in Ref. [41]. The investigated sensor configuration, which is designed to image active LEDs installed on the target, is composed of two cameras, characterized by a 2048×2048 -pixel detector imaging a $20^\circ \times 20^\circ$ FOV with resolution of 0.01°. Although the stereo baseline (30 cm) is maximized with respect to the size of the chaser, the system provides satisfying accuracy level (i.e., centimeter-level depth resolution) within a much shorter operating range (below 10 m) with respect to the VBS which also relies on actively cooperative markers.

When dealing with uncooperative spacecraft, vision-based approaches rely exclusively on monocular or stereo cameras installed on the chaser. Indeed, the VBS camera can also be used in non-cooperative mode, as demonstrated in Ref. [42] where uncooperative pose determination algorithms are tested over real images collected during the PRISMA mission. The Relative Navigation Sensor (RNS) was designed to perform 6-DOF pose estimation of a servicing platform with respect to an uncooperative target, namely the Hubble Space Telescope, by exploiting exclusively vision-based algorithms [43]. The system is composed of three monocular cameras (RNS1, RNS2 and RNS3), developed by MacDonald, Dettwiler and Associates (MDA), designed to operate at different intervals of range. Its performance was preliminary tested on ground [43], and then further assessed as an independent experiment during the Space Shuttle Mission (STS) STS-125 [44]. Two of the RNS cameras were also integrated within Argon, a test-system conceived to test capabilities to perform autonomous rendezvous in space [45]. Another example of monocular sensor whose operation is reproduced within a simulation environment to test an uncooperative pose determination algorithm can be found in Ref. [46]. Instead, some approaches involving the use of two cameras, and again tested by means of numerical simulations, are presented in Refs. [47] and [48]. However, it should be noticed that the approach presented in Ref. [47] exploits two collaborative cameras without performing any stereo calculation. The main technical specifications of these examples of monocular and binocular systems for uncooperative pose determination are collected in Tables 3 and 4, respectively.

By looking at the data collected in Table 3, it is clear that the detector must be carefully selected in order to obtain an angular resolution (i.e., the IFOV) of a few cents of a degree. Indeed, this requirement is important to identify non-cooperative features with satisfying accuracy over several tens of meters of operational range. On the other hand, one main limitation of binocular configurations is that the baseline is selected to provide satisfying depth resolution in a small range of relative distances, e.g., below 5 m in Ref. [47]. Hence, stereo cameras appear to be less flexible and less convenient than monocular cameras in terms of operational distances, weight, power consumption and required processing power. However, they can be successfully exploited for maneuvers, e.g., final approach, involving very short target/chaser distances.

Table 2

Main characteristic of spaceborne monocular systems for cooperative pose determination.

System (year of mission launch)	Mission (Developer)	Operational range	FOV	Data rate	Weight	Power consumption
PXS (1997)	ETS-VII (NASDA)	below 3 m [27]	16° × 12° (conic) [27]	2 Hz [27]	N/F	N/F
AVGS (2007)	OE (MSFC)	below 150 m [38]	16° × 16° (conic) [32]	5 Hz [32]	9 Kg [38]	14 W stand-by mode 30 W tracking mode [38]
VBS (2010)	PRISMA (DTU)	below 50 m [37]	22° × 16° (rectangular) [39]	1 Hz [37]	0.425 Kg [40]	3 W [40]

Table 3

Examples of monocular cameras for uncooperative spacecraft pose determination.

Monocular sensor	Pixel size (μm)		Pixel number		FOV (°)		IFOV (°)		Focal length (mm)
	hor.	vert.	hor.	vert.	hor.	vert.	hor.	vert.	
VBS camera [42]	8.6	8.3	752	582	22.3	16.8	0.03	0.03	16.4
RNS3 camera [43]	6.4	6.4	1024	1024	23	23	0.02	0.02	16
Simulated camera [46]	7.4	7.4	640	480	49.1	37.8	0.08	0.08	5.2

Table 4

Examples of stereo cameras for uncooperative spacecraft pose determination.

Stereo cameras	Pixel size (μm)		Pixel number		FOV (°)		IFOV (°)		Focal length (mm)	Stereo baseline (m)
	hor.	vert.	hor.	vert.	hor.	vert.	hor.	vert.		
Simulated cameras [47]	27.5	27.5	512	512	22.3	16.8	0.11	0.11	13.5	0.4
Simulated cameras [48]	3.2	3.2	2048	2048	14.9	14.9	0.01	0.01	25	0.5

2.2. LIDAR

As regards the technological solutions, LIDARs can be ranked into three main groups: scanning devices (e.g., single-point, slit or pattern-projection scanners) [49], sensors based on detector arrays [50], and spatial light modulators [51]. Scanning LIDARs have typically one detector and they are able to move the direction of a single narrow laser beam by means of lenses, mirrors or other devices moved by high-speed and high-precision galvanometers. Several solutions exist for the scanning pattern ranging from the classical raster scan to more complicated patterns, e.g., Lissajous, rosette and spirals, which allow covering the FOV faster since they do not require stopping the scanning mirrors at the end of a line [52]. These sensors are relatively easy to calibrate since the user only needs to be concerned about the light sensitivity and timing for one detector. Nevertheless, the use of scanning devices involves issues which should be considered. First, the presence of moving parts can be a source of hardware failure. Second, the achievable frame rate is limited by the time needed to mechanically scan the FOV. Third, and most importantly, the relative motion between the target and the sensor during the scan time can cause undesirable motion blur effects in the acquired point cloud. The second major class of LIDARs contains those instruments which illuminate the entire scene with a single broad laser beam and use a detector array to detect and analyze the echoes back-scattered in the pixel direction. They do not have moving parts and their images are not affected by noticeable motion blur. However, scannerless LIDARs have more detectors, thus being more challenging to calibrate, and the fabrication limit on the size of the detector arrays restrains the size of the point cloud, thus getting poor and fixed spatial resolution. Finally, spatial light modulators basically illuminate portions of the scene according to a pre-defined pattern and process the radiation back-scattered toward a single detector by using compressed sensing algorithms [53]. While the former two categories were already used for relative navigation on board of flown spacecraft and for flight experiments, spatial light modulators are innovative instruments, still under development, and far from practical space application [54].

Regarding the type of laser source, LIDARs can be divided into pulsed and continuous-wave (CW) systems. CW LIDARs compute distances by measuring the phase difference between an amplitude-modulated (AM, more frequently used) or a frequency modulated (FM) emitted signal (the reference signal) and the reflected echo, thus exploiting the heterodyne principle. The major issue is that their applicability is limited to close-range applications by the phase integer ambiguity problem [54]. CW LIDARs can also exploit the triangulation method as measurement principle [55]. This procedure can ensure millimeter or even sub-millimeter error. However, the accuracy decreases as the inverse of the square root of the distance [56]. For this reason, CW LIDARs based on triangulation are typically used in close-range applications, e.g., below 15 m. Pulsed LIDARs are direct-detection systems which compute distances by measuring the time delay between transmitted and received laser pulses, typically exploiting Gallium Arsenide Time-Of-Flight (TOF) counters. These systems, also known as TOF LIDARs, use an incoherent laser source, thus representing the most convenient choice when the interval of operating distances is very large, i.e., from a few meters to a few kilometers, since they can provide constant accuracy in the range measurement as the target distance varies. However, the range accuracy is limited since it depends on the achievable time resolution. For instance, a time resolution of tens of picoseconds is needed to get centimeter-level accuracy, thus requiring complicated and expensive electronics. Scannerless systems relying on a pulsed laser source are typically indicated as flash LIDARs. A summary of this discussion is provided in Fig. 2.

A survey of active laser-based instruments used or under development for spaceborne pose determination and relative navigation purposes is provided below. The focus is on their main technical specifications and performance. The Laser Range Scanner (LARS), developed by the Canadian Space Agency (CSA), is a scanning LIDAR capable of surface imaging, as well as ranging and high frequency tracking of cooperative targets [56–58]. It was exploited during STS-52 to test capabilities of 3D inspection and surface-reflectance monitoring of the ISS. LARS can operate at short-range exploiting a CW laser source and the triangulation principle, as well as at long-range measuring the TOF of pulsed laser shots.

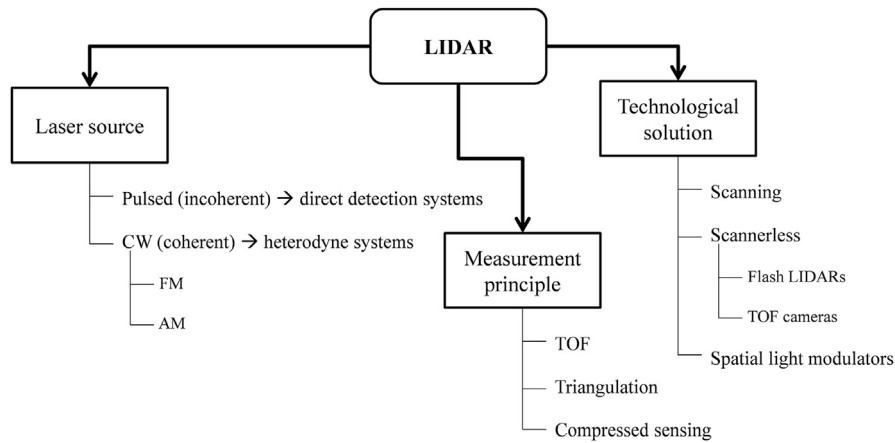


Fig. 2. Taxonomy of LIDARs.

LARS has an extremely wide dynamic range of intensities, thus being suitable for operating in space, where surfaces can generate high contrasts and specular reflections. The Laser Dynamic Range Imager (LDRI), developed by the Sandia National Laboratories (SNL), is a scannerless LIDAR qualified for space flight applications including structure characterization and proximity navigation. It was used to remotely measure vibration of the ISS structure and to determine its structural modal frequencies and amplitudes during STS-97 [59]. The LDRI illuminates the scene with a CW-AM diffuse source, and images the reflected intensity on a 640×480 -pixel detector. The Laser Camera System (LCS), developed by Neptec, is a triangulation scanning LIDAR designed to carry out on-orbit inspection and relative navigation [60,61]. Specifically, it was used during STS-118 and STS-122 to scan the Shuttle's thermal protection system at stand-off ranges below 3 m, thus being able to detect cracks less than 1 mm-thick and facilitate clearance of the vehicle for landing. The LCS can operate in "Imaging Mode" for inspection tasks, as well as in "Centroid Acquisition Mode" to identify and track the position of discrete objects (e.g., retro-reflectors on the target surface) by exploiting Lissajous scanning patterns. The Laser Mapper (LAMP), developed by the Jet Propulsion Laboratory (JPL), is a TOF scanning LIDAR designed to tackle several guidance and navigation applications in space, including rendezvous or docking with another spacecraft in Earth orbit [62,63]. It operated as the primary relative navigation sensor during the demonstration mission XSS-11 to carry out the Autonomous Rendezvous Experiment (ARX). LAMP is able to detect both cooperative targets (e.g., retro-reflectors) and uncooperative Lambertian surfaces up to a distance of 5 km and 2.5 km, respectively. The Rendezvous and Docking Sensor (RVS) is a TOF scanning LIDAR developed by Jena-Optronik to support and control the automated docking of unmanned transfer vehicles with the ISS [64]. It was first used on board of the first Automatic Transfer Vehicle (ATV-001), although the RVS prototype had already been qualified in orbit during STS-84 and STS-86 for the docking maneuver to the MIR space station. New generations of this sensor, namely the RVS-3000 and RVS-3000 3D (which are both lighter and less power consuming than the basic RVS) were recently developed by Jena-Optronik for rendezvous and docking applications. While RVS-3000 flew as an experiment on the ATV-005 vehicle, RVS 3000 3D represents a further advancement being able to autonomously identify and detect targets by measuring high-resolution point clouds. The Triangulation LIDAR sensor (TRIDAR) was also developed by Neptec. As in the LARS case, TRIDAR combines in the same unit a short-range, high precision and auto-synchronous triangulation sensor (a modified version of the LCS), with a mid-to-long-range TOF LIDAR [55]. The two sensors share the same optical path and control electronics resulting in a compact package with multi-range and multi-role capabilities. The data collected by the system can be used to perform real-time tracking of the target pose. It was used during STS-128, STS-131 and STS-135. TRIDAR represents the first active vision system,

tested in space, able to automatically acquire and track a space target in an uncooperative way. Finally, three examples of flash LIDARs for on-orbit applications are given by the DragonEye and GoldenEye Space Cameras developed by Advanced Scientific Concepts Inc. (ASC Inc.) [65], and the Vision Navigation System developed by Ball Aerospace [66]. DragonEye and GoldenEye are characterized by an array of 128×128 pixel, while the VNS detector is composed of 256×256 pixels. Hence, these systems can acquire point clouds comprising approximately 2×16300 and 2×65500 range and intensity data. A DragonEye prototype successfully flew during STS-127 and STS-133 for performance assessment of autonomous rendezvous and docking operations, while GoldenEye is foreseen to be used for relative navigation with respect to an asteroid in the framework of the OSIRIS-REX Asteroid Sample Return mission [67]. A VNS prototype was installed on board of the Space Shuttle during STS-134 to preliminary assess its performance, while it will be the primary cooperative relative navigation sensor for the Orion Multi-Purpose Crew Vehicle [68]. The possibility to exploit VNS as an uncooperative pose determination sensor was also tested on ground by integrating it within the Argon system [45]. A summary of the

Table 5a
Survey of currently existing and under development spaceborne LIDAR systems.

System (developer)	Operational mode	Technology & measurement principle	Operational range (m)	Documented range accuracy
LARS (CSA)	cooperative	Scanning CW Triangulation Pulsed TOF	0.5–10 10–10000	sub-mm 3 cm
LCS (Neptec)	cooperative	Scanning CW Triangulation	1–10	0.1 mm - 5 mm (1σ)
LAMP (JPL)	non-cooperative	Scanning Pulsed	<5000	10 cm (bias)
	non-cooperative	TOF	<2500	2.6 cm (3σ)
RVS (Jena-Optronik)	cooperative	Scanning Pulsed TOF	1–1500	0.01 m - 0.5 m (bias) 0.01 m - 0.1 m (3σ)
RVS-3000 (Jena-Optronik)	cooperative	Scanning Pulsed	1–1500	N/F
	non-cooperative	TOF	1–100	
TRIDAR (Neptec)	non-cooperative	Scanning CW Triangulation Pulsed TOF	0.5–2000	N/F
LDRI (SNL)	non-cooperative	Scannerless CW AM	<45	0.25 cm
DragonEye (ASC Inc.)	non-cooperative	Scannerless Pulsed TOF	<1500	10 cm (bias) 15 cm (3σ)
GoldenEye (ASC Inc.)	non-cooperative	Scannerless Pulsed TOF	<3000	10 cm (bias) 15 cm (3σ)
VNS (Ball Aerospace)	Cooperative (potentially non-cooperative)	Scannerless Pulsed TOF	<5000	10 - 20 cm (3σ at 10 m)

information about the above-listed sensors is given in Table 5(a) and Table 5(b).

The systems designed to deal with cooperative targets are characterized by large intervals of operating distances and high range accuracy. Indeed, this depends on the higher intensity of the radiation back-scattered toward the detectors' surface if the target is covered by retro-reflectors, with respect to echoes received from uncooperative (Lambertian) surfaces. Instead, sensors designed for non-cooperative applications can provide satisfying accuracy levels up to smaller distances. In terms of range accuracy, scanning systems are better performing than flash LIDARs, which, however, are lighter and slightly less power consuming. Since the FOV is imaged instantaneously by all the pixels composing the detector, scannerless systems can also provide more frequent measurements but they are characterized by poorer resolution. This latter issue may be solved by fusing data collected by LIDARs and high-resolution cameras, thus gathering complementary benefits, namely the 3D dimensional information coupled with the resolution of advanced 2D imagers [66]. Most of the surveyed instruments operate at a wavelength around 1500 nm which is not only eye-safe but also corresponds to a fall of the solar irradiance level. The size of their FOV is comparable in both azimuth and elevation, namely around 30° and 40°. This choice allows reducing the amount of data to be processed and, consequently, the computational load of pose determination algorithms. However, it also determines the necessity to improve their performance in terms of computational efficiency to avoid losing track of the target. Indeed, this phenomenon can occur when dealing with uncooperative debris characterized by fast tumbling motion.

3. Cooperative pose determination

The logic behind cooperative pose determination algorithms is that the relative attitude and position of a Target Reference Frame (TRF) with respect to a Sensor Reference Frame (SRF) can be estimated relying on a limited number of features that are extracted from the acquired datasets. Specifically, they correspond to passive/active artificial markers which typically produce more distinguishable echoes with respect to natural features (e.g., corners), and whose 3D position in TRF is known since they are purposely mounted on the target surface. On this basis, the pose determination process has to include techniques suitable for identifying the detected markers, which means to correctly match them within a catalogue stored on board. Depending on the selected EO sensors, cooperative pose determination algorithms can be either monocular or 3D.

Monocular images are firstly processed to extract the LOS of the markers' centroid in SRF. This can be done by using quite standard image

processing algorithms which typically include image segmentation and feature detection steps. Of course, the choice of the most suitable technique depends on the markers' geometry and configuration. Some examples are shown in sub-section 3.1 which focuses on spaceborne applications. The extracted features are then matched to the real markers (by exploiting information on either their individual shape or their mutual geometry), thus leading to a certain number (n) of 2D-to-3D point correspondences. Given n matches between real world and image points, the process of estimating the relative position and orientation of a camera with respect to the observed scene (e.g., the target), which is generally known as extrinsic camera calibration [69], can be tackled by solving the Perspective- n -Point (PnP) problem. This task, firstly introduced by Fischler and Bolles to determine the 3D position in SRF of the image points if their corresponding 3D position in a real-world coordinate system (e.g., the TRF) is already known [70], has recently assumed a more general meaning within the interested research community, i.e., equivalent to pose determination. Indeed, once the 3D positions of a set of target points is known in both SRF and TRF, the estimation of the 6-DOF pose parameters is straightforward. This point is clarified later in this section. Provided that at least three non-collinear real-world points are imaged and are not aligned with the camera optical center, the PnP problem admits a finite number of solutions [71], as analyzed in detail in Table 6.

Recently, strong efforts have been addressed to find PnP-solutions which are simultaneously accurate, fast and robust to outliers [75–77]. This is important to discard inaccurate matches caused by false detections of markers. This may occur in the space environment due to the peculiar visibility conditions.

Once the markers' 3D position in SRF is known, monocular and 3D approaches do not have conceptual differences, since they both consists in determining the relative attitude and position between two reference frames given a set of n 3D-to-3D point correspondences. Of course, 3D approaches still require a preliminary processing step of the acquired 3D data to extract the cooperative markers and carry out the matching function. These aspects are highlighted in sub-section 3.1 with reference to spaceborne applications.

If n is three, at least two directions (identified by the marker's relative position vectors) can be computed in both SRF and TRF. This information can be used to reconstruct the relative rotation matrix by implementing TRIAD [78], i.e., a deterministic method for which many variants exist [79–81]. If n is larger than three, potentially $(n^2-n)/2$ distinct unit vectors can be estimated in both the reference frames. These multiple vector observations can be exploited for relative attitude calculation using probabilistic approaches like the QUEST algorithm [78], which has also been subjected to several developments [82,83]. Given the relative rotation matrix, the position of a certain marker with respect to the TRF and SRF origins can be expressed in the same reference frame, so that their difference represents an estimate of the relative translation. It is worth outlining that the pose estimation process can be carried out exploiting directly the 3D position information of the markers. Specifically, the 6-DOF pose parameters can be derived by looking for the optimal transformation necessary to align two patterns of 3D points. This

Table 5b
Survey of currently existing and under development spaceborne LIDAR systems.

System (developer)	Mass (kg)	Power Consumption (W)	Operating wavelength (nm)	FOV (°)	Data rate (Hz)
LARS (CSA)	6	6	1540	30 × 30	N/F
LCS (Neptec)	12.1	75 (max)	1500	30 × 30	5
LAMP (JPL)	5.9	35	1064	10 × 10	2
RVS (Jena-Optronik)	14.5	40–70	910	40 × 40	N/F
RVS-3000 (Jena-Optronik)	8	30–50	1500	40 × 40	1–4
TRIDAR (Neptec)	15.9	70	1540 (TOF) 1400 (triangulation)	30 × 30	5–10
LDRI (SNL)	2.3	37	807	40 × 40	7.5
DragonEye (ASC Inc.)	3	35	1570	45 × 45	5–30
GoldenEye (ASC Inc.)	6.5	50	1570	45 × 45	5–10
VNS (Ball Aerospace)	12	30	1572	20 (conic)	30

Table 6
Characterization of the PnP problem as a function of n .

n	Upper bound of PnP ambiguous solutions [70]	Notes
3	4	Explicit criteria can be assigned to obtain from one to four solutions [72].
4	2	A unique solution can be analytically derived if the four points are coplanar [70,73,74].
5	2	A unique solution can be analytically derived unless the five points are arranged according to a set of well known critical configurations [74].
≥6	1	A unique solution can always be analytically determined [70,74].

can be done minimizing in a least-square sense a cost function defined as the sum of Euclidean distances between corresponding points [84–86], which is typically realized exploiting methods like the Singular Value Decomposition (SVD). It should be noticed that these approaches, extensively used for various robotic applications, are also suitable for uncooperative pose determination tasks for which the point correspondences are less reliable and the size of the 3D point pattern is much larger with respect to cooperative tasks. In this regard, one of the most efficient solution to this problem is given by the Iterative Closest Point (ICP) algorithm [87,88], as it is highlighted in Section 4. In conclusion, a flow diagram, summarizing the cooperative pose determination architectures relying on either monocular or 3D approaches is provided in Fig. 3. The blocks “Image Processing” and “3D Data Processing” include both the marker detection and identification tasks, while the block “3D Data Registration” is responsible for the determination of the pose parameters from n 3D-to-3D point correspondences. In the left side of Fig. 3, the role of the “PnP” block is relative to the original definition given by Fischler and Bolles in Ref. [70]. This is done to highlight the similarity between monocular and 3D architectures. However, in a more general case, the PnP can be considered as a black box which outputs directly the relative position vector and rotation matrix, given the Markers’ 3D position in TRF and their LOS in SRF.

3.1. Pose determination of cooperative spacecraft

An overview of cooperative pose determination techniques specifically tailored to spacecraft scenarios is now presented. Some of these approaches include original formulations for a relative navigation filter which are not discussed here.

3.1.1. Monocular techniques

The PXS, used in the ETS-VII mission, easily detects a set of passive cooperative markers since it illuminates the target docking interface with active LEDs from a very short distance. The identification is done exploiting the distinct size of the round-shaped markers, which are shown in Fig. 4 (extracted from Fig. 5 in Ref. [27]).

Pose estimation is carried out considering a couple of symmetric

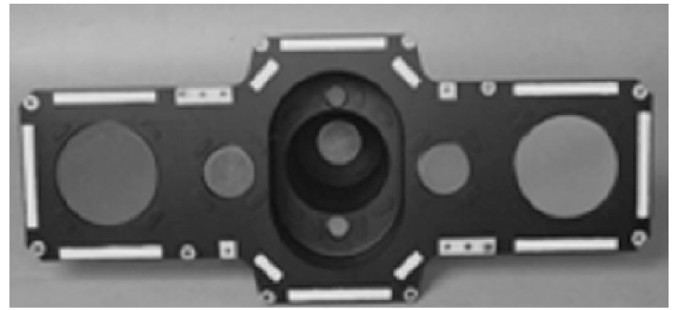


Fig. 4. PXS markers [27].

coplanar markers (the size of the selected couple reduces at lower distance) plus the non-coplanar marker. Concerning the relative attitude, pitch and yaw rotations are computed by evaluating the position of the non-coplanar central marker, while roll is computed by detecting the rotation of the baseline vector formed by one of the three sets of symmetrical markers. The relative position components can be estimated by detecting the variation of the baseline length (longitudinal axis) and the displacements on the focal plane of the baseline center (transverse axes) [27]. During the ETS-VII mission, PXS attained centimetric accuracy in relative position and tenths of degree of accuracy in relative attitude estimation [27].

The AVGS, operated during the OE mission, extracts the target’s CCRs by means of a background-subtraction method [89]. It consists in subtracting two consecutive images in which the target has been illuminated by the 808-nm and 845-nm LEDs, respectively. Since, the CCRs are equipped with a lens which filters the 808-nm wavelength, the first frame contains only the background while the second frame includes potentially valid spots. After this step, the CCRs are distinguished from other spots exploiting the knowledge of their size and mutual geometry. Finally, the pose is reconstructed by exploiting an Inverse Perspective algorithm whose detailed description can be found in an internal memo of the MSFC [90]. Qualitative information about the adopted pose estimation technique can also be found in Refs. [31,89]. During OE, AVGS

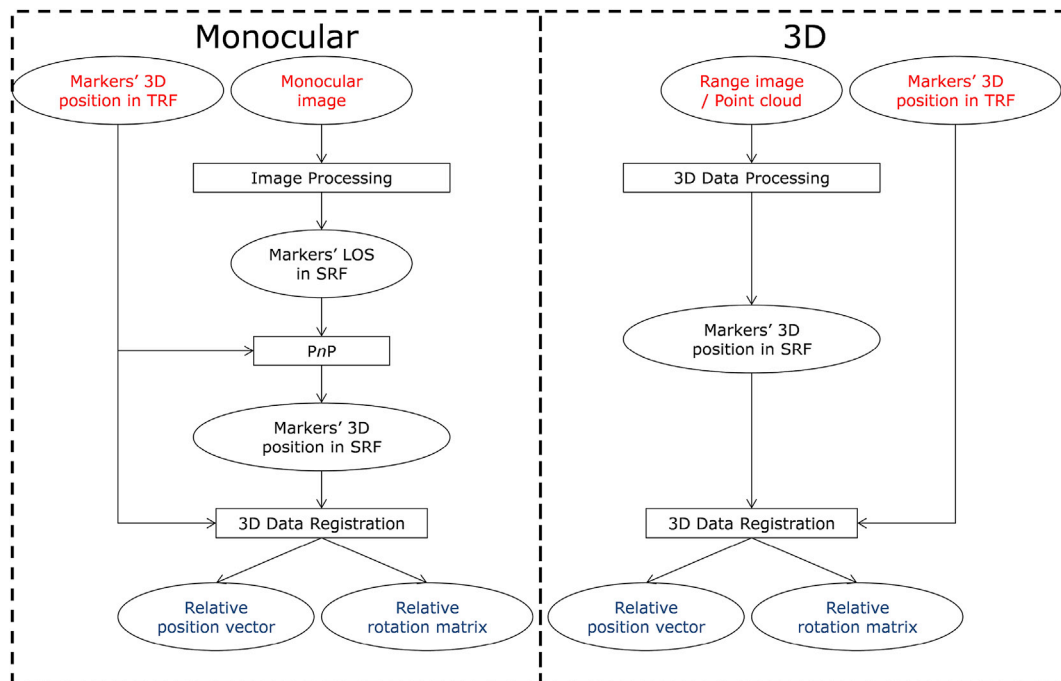


Fig. 3. Cooperative pose determination architectures for the monocular (left) and 3D (right) approaches. The input and final output are highlighted in red and blue, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

estimated the pose below 10 m using the short-range CCRs, and above 30 m (up to 150 m) using the long-range CCRs. Instead, it supplied a dual solution (imaging both the two sets of CCRs) in the intermediate range (dual-track mode). Details about the accuracy levels attained by the AVGS at variable operational ranges can be found in Refs. [32,91], including a comparison with the OS requirements. Since the docking mechanism had a very tight tolerance (much finer than the resolution of the AVGS camera), a set of “true” pose data was available to evaluate AVGS accuracy during the static, mated OE configurations (the true range was 1.22 m). Specifically, the statistical analysis of results relative to the mated data samples showed a relative position accuracy of 0.75 mm (1σ) which resulted to be one order of magnitude better than the requirement (12 mm) [32].

With regards to the PRISMA mission, the processing of the VBS images to detect the LEDs mounted on the target surfaces (five cooperative markers are located on each panel) is less prone to generate outliers with respect to the extraction of CCRs performed by the previously mentioned systems. Indeed, active makers produce brighter returns on the image plane than any other object in the scene, while CCRs may more easily be mistaken for other objects backscattering large amounts of energy when illuminated by the active sources on the chaser. If at least four LEDs are extracted, the pose is estimated by solving the P4P problem, as proposed in Ref. [73]. The determination of correct 2D-to-3D point correspondences is done by exploiting the unique pattern according to which the LEDs are located on the panels. If more than four LEDs are detected, multiple pose solutions are computed considering the potential matches between the real pattern and all the combination of four marker-centroids on the image plane. The ambiguity is solved by selecting the set of estimated pose parameters which provide the best fitting between real markers and their image projection [35]. During PRISMA, VBS provided 1-cm accuracy at 10-m range, while sub-millimeter and sub-degree accuracy was achieved in laboratory conditions at the same distance. Although the positioning accuracy was shown to change as a function of the relative distance, the adopted algorithm demonstrated good robustness against low visibility conditions and in presence of bright celestial objects in the field of view.

Back from their AVGS development and testing experience, Becker et al. [92] designed the Smartphone Video Guidance Sensor to enable proximity operations and formation flying capabilities for small space platforms (e.g., CubeSats). Specifically, it is a miniature sensor which relies exclusively on the camera, the flash and the processing resources of a commercial, off-the-shelf, Android-based Smartphone to carry out 6-DOF pose determination of a cooperative space target equipped with CCRs mounted according to a known pattern. Like for the AVGS short-range markers, the pattern is composed of four CCRs. A potential mounting solution consists in locating three coplanar CCRs on one of the surfaces of a cubesat. The remaining marker is mounted with an offset in the corresponding normal direction (see Fig. 1 in Ref. [92]). For each detected frame, a binary image is created by subtracting a specified brightness threshold value. This image will contain several spots representing potential candidates for matching with the target CCRs. The identification task is accomplished by finding the best match between the real pattern and all the possible combination of four spots. Finally, the pose vector is reconstructed using a customized implementation of the collinearity equations [92]. A drawback of the proposed approach is that it relies on the prior knowledge of a tentative pose solution. Performance assessment was conducted experimentally within a ground test-bed available at MSFC. The SVGS was mounted on a structure which allowed varying the attitude (3 rotational DOFs) as well as the range to a fixed target (1 translational DOF). By using a Galaxy-Nexus™, the proposed system provided pose estimates from 5 m to 30 m of relative distance, ensuring a worst-case accuracy (at the maximum range) below 1 m and around 1° for the relative position and attitude, respectively. On orbit demonstrations of SVGS for cubesat applications are foreseen as future developments.

Another approach for cooperative pose determination, specifically

tailored for OOS applications involving small satellites, is proposed in Ref. [93]. The authors integrate in an original manner multiple existing techniques conceived for the different steps of the pose determination process, including a relative navigation filtering scheme. Specific cooperative markers, namely the concentric contrasting circles [94], are selected, since they are passive and are purposely designed to minimize their size on the target vehicle. Also, their geometric properties ensure critical advantages in finding image/real-world point correspondences since the centroids of the circles remain co-located under rotations and translations. Additionally, the area ratio between the outer ring and inner circle remains constant under rotation and translation. Then, the pose is estimated by means of a globally convergent, non-linear, iterative algorithm which exploits a solution to the P4P problem [95]. Performance of this approach was characterized by means of experimental tests realized on ground exploiting the SPHERES satellites as target and chaser vehicles [96]. Centimeter- and degree-level accuracy was demonstrated in relative position and attitude, respectively.

An original solution for the selection of cooperative markers is proposed in Ref. [97]. This work describes and calibrates, from a metrological point of view, a vision-based system conceived for pose determination of a cooperative spacecraft. Specifically, the markers have a square shape and each of them includes a predetermined number of internal dots. The four corners of each marker are used to compute precisely the marker centroid thus solving the marker-detection problem. Instead, the number of internal dots are used to determine correct matches between the 2D markers-centroids on the image plane and their real 3D position in TRF. The relative position and attitude parameters are estimated exploiting a solution to the P3P problem originally conceived in Ref. [98] for visual odometry applications. However, as shown in Ref. [70], the P3P problem contains an unresolved ambiguity which, in this case, leads to four pose solutions. The ambiguity is resolved by projecting a fourth marker on the image plane according to the four pose solutions and selecting the relative position and attitude parameters for which the re-projection error with respect to another marker-centroid is minimum. The obtained pose is iteratively re-computed using a RANSAC scheme to eliminate false or inaccurate matches and to identify the inliers. This is done since these points can be used to further refine the pose by optimizing a non-linear cost function (the squared sum of the inliers re-projection error) in a least-squares sense. Performance assessment was carried out within an experimental setup designed to modulate 2 DOFs, namely the target distance from the camera (22 steps between 1.5 m and 2.55 m) and the target rotation around a vertical axis (45 steps from 0° to 90°), thus leading to 990 analyzed configurations. The average accuracy was around 1 cm and 1° in relative position and attitude, respectively. Based on the attained results, this approach appears to work better if the optical axis is squinted enough (e.g., 20° – 30°) with respect to the normal direction to the target surface.

3.1.2. 3D techniques

Moving on to 3D approaches, performance of the pose determination techniques based on stereo measurements illustrated in Ref. [41] is assessed by means of numerical simulations. However, in this work, the problem of detection and identification of cooperative markers is not addressed. Instead, the relative dynamics corresponding to a typical target monitoring scenario (the chaser is on a relative circular orbit) is realistically reproduced. TRIAD and QUEST solutions for the estimation of the relative rotation matrix are compared, while the relative position vector is obtained as the average of the vector differences between the computed positions of the same markers in SRF and TRF. The proposed algorithm ensures sub-centimetric and sub-degree accuracy in relative position and attitude, respectively. Also, it allows demonstrating that QUEST performance is approximately 50% better than three different implementations of the TRIAD algorithm (obtained using different couples of observation vectors).

Regarding active sensors, reference [99] provides a detailed discussion on how to use Flash LIDARs for relative navigation in close-

proximity (below 15 m) to a cooperative target spacecraft. The entire procedure, which leads to the estimate of the target relative position and attitude, is analyzed, starting from the processing of raw data (which include 3D position and intensity information for each pixel). The first step is the extraction of a set of candidate reflectors by implementing a procedure based on morphology. This procedure includes multiple thresholding and filtering operations to be applied to the LIDAR intensity image. Then, the reflector identification process is carried out by a graph-matching algorithm for which a rich literature exists [100]. In this regard, an approach to design 3D point patterns which are uniquely distinguishable in LIDAR data, even in absence of an a priori pose estimate, can be found in Ref. [101]. Apart from the specific proposed solution, reference [99] provides a list of requirements that algorithms conceived for reflector detection and identification need to satisfy to be reliable for space applications. Finally, three strategies for pose estimation are described and compared in terms of the achieved accuracy, robustness/stability, speed, and complexity. The first technique is the SVD-based algorithm proposed in Ref. [95] by Haralick to solve the P4P problem and modified to use directly the data collected by a 3D LIDAR. The remaining methods are variants of the optimal linear attitude and translation estimator (OLATE) algorithm [102], which use the Cayley transform to parameterize the orthogonal rotation matrix in terms of the Rodriguez parameters, thus obtaining a linear formulation of the 3D point pattern aligning problem. Numerical simulations are reported considering a target with five reflectors at distance from 2 m to 20 m. They show that the SVD-based algorithm outperforms the OLATE-based ones in accuracy, stability and speed, having an accuracy around 1° and 1 cm in relative attitude and position, respectively.

3.1.3. Summary

Some interesting conclusions may be drawn at the end of this survey, that are valid for both monocular and 3D approaches for pose determination of cooperative spacecraft.

- 1) The most critical task within the overall pose determination architecture is to correctly detect and identify the cooperative markers.
- 2) The accomplishment of this task is strictly related to the design of easily recognizable markers, which need to be located according to unique patterns.
- 3) Both the individual markers and the pattern need to be as invariant as possible to illumination, relative position and attitude conditions.
- 4) Performance assessment is carried out by means of on-orbit demonstration, experimental tests, and/or numerical simulations.

- 5) These analyses demonstrate that most approaches (both monocular and 3D) can provide centimetric and sub-degree accuracy in relative position and attitude, respectively, operating at target/chaser distances of several tens of meters (thus being compliant with the requirements of close-proximity operations).

4. Uncooperative pose determination: known targets

When dealing with uncooperative targets, the general architecture of the pose determination process involves two different steps, namely acquisition and tracking. Pose acquisition is performed when the first dataset is provided by the adopted sensor and no a-priori information about the target relative position and attitude is available. Pose tracking means updating the pose parameters, as soon as new datasets are acquired, using the knowledge of pose estimates at one or more previous time instants. This distinction is not relevant to cooperative targets since the necessity to detect and identify the cooperative markers excludes that substantial differences arise in the complexity of the pose determination task depending on the considered time instant.

The uncooperative pose determination process may include an additional step of pose refinement to improve accuracy and computational efficiency of pose tracking. This step can be used either to both update the final pose estimate and re-initialize the tracking algorithm for the subsequent measurement time [103], or just to accomplish this latter task [104]. Another variant includes two different refinement procedures implemented after both pose acquisition and tracking [42]. These three examples of pose determination architectures (designed in the context of relative navigation with respect to an uncooperative space object) are depicted in Fig. 5.

In this section, the uncooperative target is assumed to be known, which means that either detailed information about its geometry, or at least a simplified geometric model is available. Under this assumption, uncooperative pose determination can be handled by model-based algorithms. Specifically, they allow estimating the pose vector by matching data extracted from measurements of monocular/3D sensors with similar information derived from the target model stored on board. This task is typically accomplished by using feature-based approaches which rely on the capability of extracting natural geometric features from the acquired dataset, such as corners, lines, curves, contours, or 2D/3D point descriptors. Of course, different techniques can be used depending on their role within the pose determination process (acquisition or tracking), as well as on the type of available data (monocular or 3D).

If monocular sensors are used, feature-based algorithms for pose acquisition can rely on either PnP solvers or Template Matching (TM)

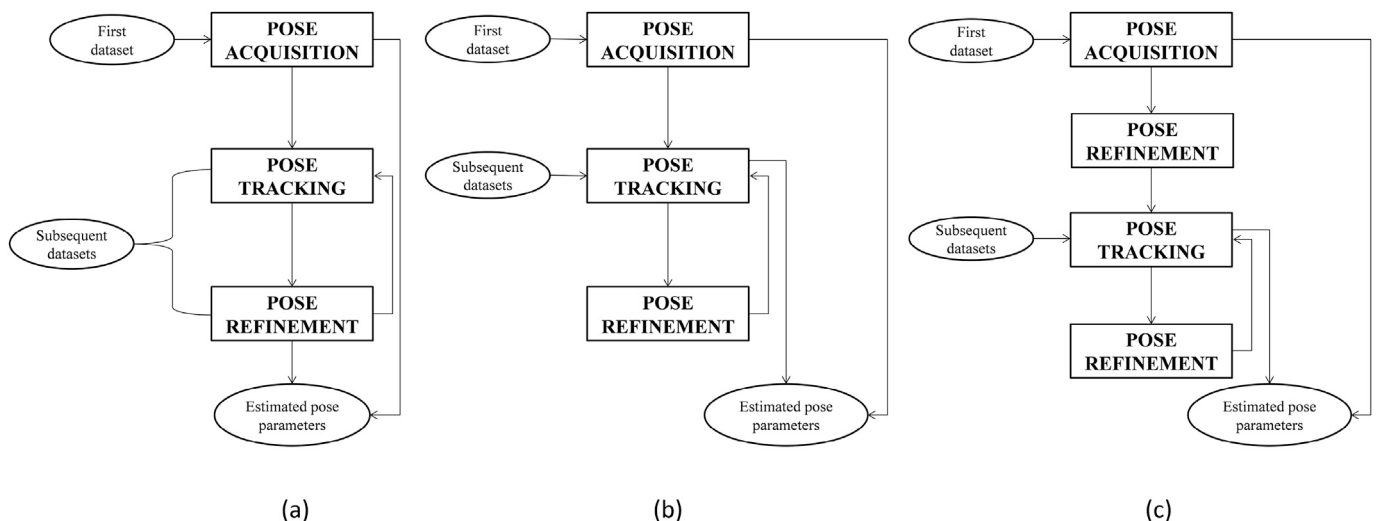


Fig. 5. Logical schemes for different uncooperative pose determination architectures. (a) [103], (b) [104], (c) [42].

approaches. As regards the former category, the issue of finding correct correspondences between the features extracted from the acquired image and the target model is extremely challenging due to the absence of fiducial markers on the target surface. An iterative, but linear, solution to the problem of searching for image-to-model correspondences and pose estimation is provided by the SoftPOSIT algorithm [105]. First, it uses the soft-assign technique to determine image/model point correspondences [106,107]. Afterwards, the POSIT (Pose from Orthography and Scaling with Iterations) algorithm is used to iteratively estimate the pose [108]. Specifically, it uses the POS (Pose from Orthography and Scaling) algorithm to approximate the perspective projection with a scaled orthographic projection, thus finding a coarse estimate of the pose parameters. Hence, the approximated pose is used to compute better scaled orthographic projections of the feature points. This process is iterated until convergence. The SoftPOSIT applicability can be extended to the case in which lines are detected in the image instead of points [109]. This is important since lines are typically more stable than points, i.e., more invariant to changes in lightning conditions, and less likely to be produced by clutter and noise. Apart from POSIT, some examples of different PnP solvers have been presented in Refs. [75–77]. Instead, alternative solutions to soft-assign for the correspondence problem are given by RANSAC [70] or rely on complex 2D image descriptors [110]. Interested readers can find in Ref. [111] a more detailed overview of feature-based methods for pose acquisition from monocular images, that rely on points, lines, more complicated primitives (such as sets of collinear points, spheres, cylinders) or even combination of multiple features (thus leading to the concept of perceptual grouping [112]).

TM is a term used in computer vision to indicate the process of searching a monocular image for specific features and/or image sections which can be matched to an assigned template [113]. The template can have the same size as the available dataset, or it can occupy a limited area of it, while the matching function is carried out by exploiting a correlation approach, e.g., based on the sum of absolute differences [114], the normalized cross correlation [115], or the distance transform [116]. With respect to pose determination, TM requires the generation of a database of templates by sampling the 6-DOF pose space. Specifically, each sample set of relative position and attitude parameters is used to generate a template-image of the target. On this basis, the selected correlation function is computed to evaluate the degree of similarity between each template and the acquired image. Hence, the best-correlation template provides the unknown pose vector. The TM logical scheme is summarized in Fig. 6. An example of TM algorithm, applied to extract the initial target pose from monocular data and based on the extraction and matching of the target silhouette (i.e., the contour), can be found

in Ref. [117].

Once the initial pose estimate is available, the relative position and attitude parameters corresponding to subsequent frames (pose tracking) can be determined using the procedure indicated below.

- 1) The target model is represented as a sparse set of features, typically lines (e.g., edges) [118,119] or curves (e.g., circles and ellipses) [120].
- 2) The same features are extracted in the acquired image.
- 3) The model is projected onto the image plane according to the initial pose estimate.
- 4) The actual pose is obtained as the transformation that provides the best fit (alignment) between projected and image features. This latter step is typically performed by optimizing a purposely defined cost function, e.g., in a least squares sense.

When dealing with monocular images, an alternative solution to feature-based approaches is given by appearance-based methods. These techniques analyze the shape and texture of the acquired datasets by exploiting either the Active Appearance Models (AAM) [121] or the Principle Component Analysis (PCA) [122]. Interesting examples for AAM-based and PCA-based methods are given by Refs. [123] and [124], respectively. Despite being tailored for pose estimation or recognition of objects having complex articulated shapes, e.g., human hands or faces, appearance-based algorithms are less suitable for applications in space where the scene can be poorly textured. Also, they always need an initial estimate of the pose vector as well as a pre-processing stage performed on a series of sample images to create a database.

If 3D sensors, i.e., stereo cameras or LIDARs, are used, pose acquisition can be entrusted to algorithms relying on global or local 3D feature descriptors [125–128]. Indeed, 3D feature descriptors allow matching 3D data extracted in the acquired datasets with a consistent 3D representation of the target, like for monocular applications. These methods have several non-space applications like object recognition for aerial surveillance. In this respect, a survey of methods using 3D feature descriptors for object recognition is given by Ref. [129]. A relevant alternative to feature-based approaches is given by point-based techniques which directly process 3D raw data, e.g., range images or point clouds. These solutions ensure computational time saving since no feature extraction step must be implemented. Also, they tend to be more robust than feature-based algorithm to noise, distortions, and outliers [130]. Interesting examples of point-based algorithms for pose acquisition can be derived by adapting the TM concept to deal with 3D datasets. For instance, the logical scheme of Fig. 6 is still applicable provided that adequate 3D correlation laws are selected, like the mean square distance [131] and binary correlation [132] metrics. As regards pose tracking, it can be tackled by exploiting 3D data registration methods, for which a large variety of solutions can be found in the literature [133]. These algorithms are basically able to find the best rigid transformation to align two datasets (registration) by minimizing a given cost function which measures the similarity level between corresponding elements. Among them, the ICP algorithm is probably the most commonly selected approach. Indeed, its implementation can be easily customized to the application of interest by purposely modifying one of the steps composing each single iteration of the process [134] (i.e., initialization, matching, outlier rejection, weighting, cost function selection and minimization), thus leading to ad-hoc optimal solutions. Finally, while TM approaches are typically developed for pose acquisition, an interesting exception is given by the Bounded Hough Transform (BHT) method [135]. The BHT is a point-based algorithm which limits the search for the pose solution to a discrete space obtained by sampling the continuous space around the initial relative state. A 3D template is built for each of the sampled sets of pose parameters and the correlation with the acquired datasets is computed by exploiting the classification concept (that is inherited from the Hough Transform (HT) [136]).

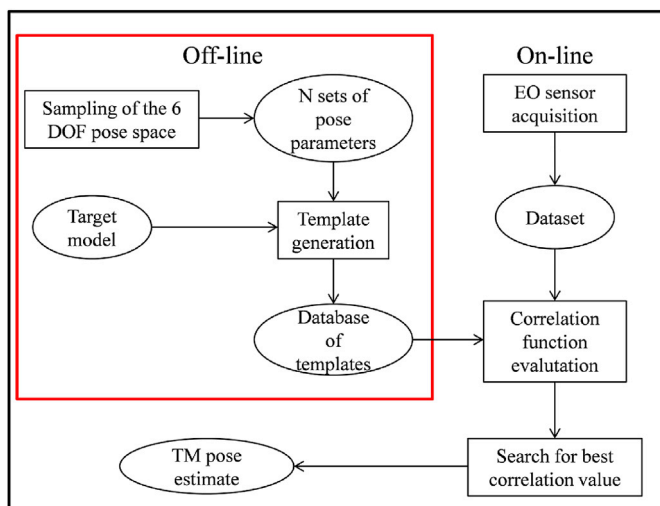


Fig. 6. Logical scheme of the classical application of the TM concept to pose determination tasks.

4.1. Pose determination of uncooperative known spacecraft

The attention is here focused on the pose determination architectures conceived for space applications involving autonomous relative navigation in close-proximity of an uncooperative target. In this context, the pose acquisition accuracy can even be coarse provided that the estimated relative position and attitude parameters fall within the field of convergence of the adopted tracking technique. Instead, the constraint on the sustainable computational cost, that is strongly related to the relative trajectory, is typically quite stringent, especially when dealing with uncontrolled tumbling objects. On the other hand, tracking algorithms must be able to run fast, i.e., in real time, and ensure high performance in terms of the estimation accuracy. In any case, the algorithms' design and implementation process should take account of the limited payload and processing resources typically available on board. This is particularly critical for small-satellite-based applications. In view of these aspects, uncooperative pose determination algorithms need dedicated solutions if used for space applications. For instance, TM techniques have to foresee strategies to accelerate the computationally heavy database search, as well as to limit the amount of stored data to be compliant with the on-board resources available for processing and storage.

Separated discussions are here proposed for monocular and 3D approaches. In both the cases, priority is given to complete architectures of pose determination. An additional hybrid approach is presented in subsection 4.1.3.

4.1.1. Monocular approaches

Astrium Satellites proposed a fully monocular-based pose determination architecture in the framework of the Debritor program [137]. Pose acquisition is composed of two steps. A silhouette TM algorithm is applied to a 2-DOF database, built off-line and organized as a hierarchical view graph [117], thus estimating two relative attitude parameters. A particle filtering framework, which is initialized by means of an object segmentation procedure, is used to estimate the remaining unknowns, namely the horizontal and vertical offsets, the scale factor and the rotation around sensor boresight [138]. The main drawback of this strategy is that the algorithm's convergence is spread over multiple frames, thus increasing the risk of losing track of the target. Pose tracking is performed by applying the edge-tracking method in Ref. [119], which is modified to improve robustness with respect to outliers. Algorithms' performance is evaluated on synthetic images of debris (provided by the Astrium rendering tool *Surrender*) and real images of the Soyuz TMA-12 spacecraft taken during its rendezvous phase with the ISS. Range and tumbling rate of the simulated targets are set variable from 10 m to 100 m and from $0^\circ/\text{s}$ to $2^\circ/\text{s}$, respectively, to be consistent with the final phase of an ADR mission. In both the cases, the achieved accuracy is around 1° in relative attitude, while the position error is approximately 1% of the true relative range [138]. As regards the computational load, the authors claim the algorithms to be still at prototype level, being able to run in real-time but only on ground.

Another approach for spaceborne pose determination of non-cooperative targets based on monocular images can be found in Ref. [42]. Acquisition is carried out by exploiting the concept of perceptual grouping [112]. As anticipated, perceptual groups (i.e., combinations of lines and points that satisfy specific geometric constraints) are more robust descriptors than individual natural features. Thus, they allow reducing the risk of ambiguous matching, which is critical in space applications due to the high-level of symmetry in the targets' geometry. According to the proposed solution, the initial pose is obtained as the one which gives the best alignment between the perceptual groups extracted from the model and the image. Pose refinement and tracking are carried out by applying a multi-dimensional Newton-Raphson method and a weighted iterative batch least-squares estimator, respectively [42]. The performance of this approach is evaluated by processing real images of the TANGO spacecraft collected during the PRISMA mission. The authors claim the tracking algorithm to

be reliable enough (accuracy around 10 cm and 10° in position and attitude, respectively) provided that the initial angular error is kept below 40° . However, they state that the major limiting factor for accuracy and reliability is in the image processing module, which is based on the HT.

The Goddard Natural Feature Image Recognition (GNFIR) and the ULTOR[®] Passive Pose and Position Engine (ULTOR) are two distinct algorithms developed and tested to perform 6-DOF pose determination of an uncooperative spacecraft by processing monocular images provided by the RNS cameras [43,44]. ULTOR can be used for both pose acquisition and tracking. It uses simulated imagery of the target to build a database offline, that is used to identify natural geometric structures of the target surface by means of correlation. Hence, a set of 2D-to-3D point correspondences are used to estimate the pose parameters solving the PnP problem. On the other hand, GNFIR is an edge-tracking algorithm. Specifically, once the edge-image is determined using a standard edge-enhancing (Sobel) filter and given an initial estimate of the pose parameters, the target relative attitude and position are updated exploiting the approach proposed by Drummond and Cipolla [118]. GNFIR can be run in real time (up to 3 Hz) on a FPGA firmware. In absence of a reliable pose estimate, GNFIR can autonomously generate an initialization (Acquisition mode) by using a prediction of the motion of both the chaser and the target [44]. The algorithm is expected to operate correctly for initial errors of 10° in relative attitude and about 10% of range in relative position. A re-initialization request is foreseen if the quality metric, i.e., the ratio between detected image- and model-edges gets below a threshold. ULTOR and GNFIR performance was assessed both on ground (within the MSFC laboratory) and in orbit [43,44]. Additional ground tests were performed for a modified version of the GNFIR integrated within the Argon system [45]. The attained results demonstrate that the main issue for both ULTOR and GNFIR algorithms is to generate an initial estimate of the pose parameters suitable for robustly tracking the target (i.e., the Hubble Space Telescope). Indeed, tracking operations were not accomplished correctly for proximity scenarios in which the target/chaser relative dynamics was particularly fast. Also, the initial acquisition time provided by ULTOR was too large (12 s - 15 s) especially considering the fast approach of the target during rendezvous maneuvers. For details, interested readers can refer to [43,44].

The above-mentioned research programs highlight that the main challenges within the uncooperative pose determination process are related to the initial acquisition step. In this respect, a comparison of different PnP solvers, namely POSIT, coplanar POSIT, EPnP, and Newton-Raphson is performed in Ref. [139]. This is done within a simulation environment designed to realistically reproduce the algorithms' input data, i.e., a set of image/target 2D-to-3D point correspondences, and considering the TANGO spacecraft as target. Several simulations are performed analyzing separately the effects of the number of point correspondences, the percentage of outlier matches, the level of pixel noise in the simulated detected features, and the inter-satellite distance. The computational runtime, the 3D points re-projection error on the image plane, and the errors in the estimated relative attitude and position are selected as performance figures of merit. The analysis of the results leads to an interesting qualitative decision matrix (see Fig. 11 in Ref. [42]) which highlights advantages and drawbacks related to the compared PnP solvers. However, the computational load needed to reach convergence at an adequately accurate pose solution is still not compatible with space implementation requirements. On this basis, the authors of [42] suggest that an optimal pose acquisition solution may be obtained implementing a PnP mode switcher, which autonomously combines the different properties of the analyzed PnP solvers to significantly accelerate the iteration process.

An optimal algorithmic solution to pose determination can be found by taking the target geometrical characteristics (shape and size) into account during the design process. For instance, a monocular-based pose acquisition algorithm specifically tailored for non-cooperative cylinder-shaped spacecraft is proposed by Liu and Hu [46]. A coarse estimate of

the symmetry axis direction and of the relative translation vector (4 DOF) is obtained by exploiting projective geometry [140] after having matched ellipses extracted from the acquired images to the ones in the model. Hence, the solution is refined by optimizing a non-linear objective function. Then, the remaining DOFs are computed by exploiting information provided by the non-symmetric components of the spacecraft, e.g., antennas and solar arrays. The accuracy of this approach is assessed by exploiting real images of the Soyuz as well as synthetic images of the same target generated by the Satellite Tool-Kit (STK). A significant issue is given by the worsening of the relative position accuracy as the rotation of the cylinder with respect to its transversal axes increases. Moreover, the method may suffer from partial surface occlusion which may deny ellipses to be detected.

Another strategy to take advantage of the precise knowledge of the target geometry is exploited within the edge-matching approach proposed by Cropp et al. [141]. Specifically, once image lines are detected with sub-pixel accuracy thanks to the HT, heuristics are used to generate a list of correspondences with the lines of a model of the target (UoSAT, a microsatellite from Surrey Space Center). Once the correspondences are found, two least-squares error functions, that satisfy specific geometric constraints [142], are minimized to compute the relative attitude and position. RANSAC is used to cope with the effect of incorrect matches. After the conclusion of the RANSAC process, a Gauss-Newton minimization is performed to refine the estimated pose. Results of numerical simulations show the capability of the algorithm to attain an average rotation error lower than 5° and an average translation error of about 2% of the target range (maximum 10 m). It should be noticed that this method requires a large computational time (around 1 min) as well as an initial guess for the relative attitude and position.

Interested readers can find details about other approaches addressed exclusively to acquisition or tracking in Refs. [103,143–147]. A summary of this review of monocular uncooperative pose determination techniques is provided in Table 7.

4.1.2. 3D approaches

As mentioned in the previous sections, 3D algorithms for pose determination in space require the use of either stereo cameras or LIDARs to provide 3D data. Although stereo systems ensure finer spatial resolution and higher frame rate than LIDARs, while being characterized by a much more limited operative range, the basic algorithmic concepts remain valid regardless of the specific source of data.

This is shown in Refs. [148,149], which present a vision system designed for spacecraft relative navigation at medium and short range by exploiting model-based 3D algorithms. Indeed, both a commercial scanning LIDAR (ILRIS-3D by Optech) and a couple of stereo cameras are used to collect data suitable for algorithms' performance assessment. Pose acquisition is done by firstly applying the PCA to find the orientation of the target main axis, thus resolving two rotational DOFs. Then, a 3D binary TM algorithm [132] is used to look for the best estimate of the remaining rotation and relative position parameters within a 4-DOF database (built off-line). This approach allows reducing the time required by the TM pose search but it has an important limitation, being best suited for space objects having an elongated shape. Pose tracking is performed by means of a customized version of the ICP algorithm [87]. Ground-based experiments are carried out within a facility including two industrial robots, one holding the instrument and the other one holding the target (a scaled replica of Radarsat-2), which follow predefined trajectories so that both acquisition and tracking modes can be tested. Further results are also obtained by means of numerical simulations. The effect on algorithms' performance of measurement noise and point cloud sparseness are analyzed. The achieved accuracy shows an asymptotic improving behavior for point clouds with less than 2000 points. Overall, the pose acquisition strategy demonstrates satisfactory performance only in case of high resolution point clouds (more than 2000 points). Below this threshold, several failures occur since less details of the target geometry are visible to solve ambiguous views. Instead, the tracking

Table 7

Survey of monocular pose determination algorithms dealing with non-cooperative space targets.

Pub. year(s) Reference(s)	Pose acquisition technique	Pose tracking technique	Performance analysis criterion
2000 [141]	Edge matching [141] + RANSAC [70] + projective geometry constraint [142]	N/A	Off-line runs on synthetic images
2006 [103]	N/A	Edge-tracker [118]	Real-time runs on real images from ground experiments
2009–2012 [43,45]	ULTOR: PnP solver [43,44]	GNFIR: edge-tracker [118]	Real-time runs on real images from both space and ground experiments
2012 [137,138]	Silhouette TM [117] + Silhouette segmentation [138] + Particle Filter [138]	Edge-tracker [119]	Off-line runs on synthetic and real images from past missions
2013 [143]	Conceptual manifold analysis [143]	N/A	Off-line runs on simulated images
2014 [144]	SIFT-based TM matching [110]	N/A	Off-line runs on real images from ground experiments
2014 [145]	N/A	Edge-tracker [118]	Off-line runs on real images from past missions
2014 [42]	Perceptual groups matching [42]	Iterative least square estimator [42]	Off-line runs on real images from past missions
2014 [46]	Ellipse matching and iterative refinement [46]	N/A	Off-line runs on real images from past missions
2015 [146]	TM based on normalized SAD [146]	N/A	Off-line runs on real images from ground experiments
2016 [139]	Comparison of PnP solvers [42,75,105]	N/A	Off-line runs on synthetic datasets
2017 [147]	SIFT/Harris-based TM matching + PnP solver	N/A	Off-line runs on real images from ground-based experiments

accuracy is below 1 cm and 0.4° in position and attitude, respectively.

These results highlight that, also for 3D approaches, the most critical step of the pose determination process is the initial acquisition of the relative position and attitude parameters. Specifically, main problems are the computational load (e.g., several seconds in Ref. [148]), the robustness against significant data sparseness, the capability to distinguish between ambiguous target poses generated by the high level of geometrical symmetry typical of space targets.

A different strategy to reduce the computational cost of pose acquisition is given by the polygonal aspect hashing (PAH) algorithm [130]. It is a point-based technique which allows accelerating the search for the pose solution within a 6-DOF database built off-line, by restraining it to those sets of parameters for which the target model surfaces are aligned to one or more polygons (composed of about 4/6 points) extracted from the input data. Given the point-correspondences for each polygon, a pose solution is computed. Performance of this method is evaluated by off-line runs on simulated and real sensor data collected by the TRIDAR sensor. The selected target is a 1:2 scale model of a Pressurized Mating Adapter (PMA) which is connected to a lateral port of a Node of the ISS. Specifically, the robustness of the PAH technique against reduction in the size of the point cloud, occlusion, and spatial resolution is tested. The algorithm is considered successful if the errors remain below 20 cm and 10° in position and attitude, respectively. High values of the success rate are obtained if the number of matched polygons is large enough (more than 4) despite the reduction in point cloud size and in sensor resolution (due to the increase in the relative distance), while the effect of occlusions starts being relevant causing a drop in performance when the target visibility goes below 50%. In case of success, the output of the PAH technique is used as initial guess for the ICP algorithm.

The idea to split the initial acquisition of the 6-DOF pose parameters

into multiple phases is also exploited by Opromolla et al. [103,131,149]. Specifically, they design and test within a realistic numerical simulator a LIDAR-based pose determination architecture, in which different approaches are proposed for acquisition and tracking. Pose acquisition is carried out to simultaneously cut down not only the processing time but also the amount of on-board stored data. Indeed, the 3D on-line TM and the 3D on-line fast TM compute the relative position with a centroiding approach, thus restraining the pose search to a 3-DOF database built on-line sampling only the attitude parameters [103,131]. Data storage saving is made possible by the fact that the database is built dynamically, meaning that each template is generated on board and directly correlated to the measured point cloud (only the correlation score is saved). The 3D PCA-based on-line TM [149] originally combines the centroiding approach in Ref. [131] with the PCA, as in Ref. [148], to estimate both the target main axis and the relative position vector by directly processing the measured point cloud. Hence, TM can be applied to a 1-DOF database built on-line. This solution allows reducing the amount of on-board data storage and ensures a lower computational burden since the number of templates (from 10 to 100) is much more limited than in previously mentioned TM techniques (e.g., [130,148]). Reference [149] also presents original and effective strategies to carry out 1) autonomous failure detection of pose acquisition 2) safe transition to the tracking stage in case of successful pose acquisition. Pose acquisition performance are evaluated by densely sampling the relative attitude space and considering a wide range of relative position conditions, and variable target geometries in terms of shape and size. This enables the identification of the regions of the pose space in which failure or undesired performance occur for a given target, thus providing guidelines to design relative trajectories which optimize the target visibility conditions for pose estimation.

Apart from the above-mentioned TM approaches, different pose acquisition solutions have been recently tested for space applications. The Oriented, Unique, and Repeatable Clustered Viewpoint Feature Histograms (OUR-CVFH) is an algorithm which associates a single descriptor to a measured point cloud (e.g., LIDAR-based) by exploiting local curvature and normal information [150]. It is part of a family of methods based on point feature histograms inherited from robotics applications, which allow both object recognition and pose estimation [151]. Specifically, a simplified point feature histogram is built by concatenating individual histograms relative to a finite number of patches (i.e., clusters of points) in which the point cloud is sub-divided. This complex descriptor associated to the point cloud is matched to a database (obtained offline by means of a training procedure which can be found in the Point Cloud Library) to estimate the pose parameters. Performance assessment is carried out within a simulation environment by reproducing rendezvous with the asteroid 25143 Itokawa. Specifically, a portion of the approach is considered to ensure that the point clouds contain enough information to reliably run OUR-CVFH. The algorithm demonstrates the capability to provide reliable initial guess to a relative navigation filter. A comparison between OUR-CVFH and a different 3D descriptor (i.e., spin images), both integrated within an Extended Kalman Filter (EKF) architecture for relative navigation, can be found in Ref. [152].

Instead, Liu et al. propose an original point-based method to initialize the ICP algorithm for uncooperative spacecraft pose determination, assuming a flash LIDAR as relative navigation sensor [153]. Specifically, the PCA is used to extract information from both the measured and model point clouds. Then, a branch-and-bound strategy [154] is exploited to perform an iterative global optimal search in the 6-DOF pose space for the set of parameters which represents the best initialization to the ICP algorithm. Performance assessment is done considering synthetic data, generated within a simulation environment, and real data, collected by a TOF camera (which provides similar data compared to a flash LIDAR, but relies on a different measurement principle). The proposed approach is able to provide an adequate initialization for target/chaser relative dis-

Regarding pose tracking, the ICP algorithm results extremely reliable for space applications even when dealing with sparse point clouds (low resolution), and with coarse initial guesses of the pose parameters (e.g., initial errors of tens of degrees in attitude and of meter level in position). Several variants of the ICP algorithm can be found in the literature which are designed to find an estimate of the optimal rigid transformation necessary to align a target-model point cloud to a dataset acquired with LIDARs [45,104,130,131,147,149,150,153,155,156] or stereo cameras [147,157]. A comparison of performance of different variants of the ICP algorithm in terms of pose estimation accuracy and computational load is presented in Ref. [158].

The BHT is used as an alternative solution to the ICP algorithm for tracking the motion of a known uncooperative target relying on either LIDAR or stereovision point clouds [159]. Three strategies, i.e., BHT-standalone, ICP-standalone, and a hybrid approach are compared on both simulated and real data. Despite comparable accuracy levels, results show that the hybrid strategy, which refines the prior BHT solution using the ICP algorithm, is the best in term of computational load.

A summary of the main reviewed approaches for pose determination of uncooperative spacecraft relying on 3D data is provided in Table 8.

4.1.3. Hybrid approaches

The idea, mentioned in Section 2, of fusing data from active LIDARs and passive cameras is exploited in Ref. [160] to develop and test a hybrid approach to pose tracking of uncooperative spacecraft. Indeed, this method aims at exploiting the complementary advantages of these active and passive sensors (i.e., 3D position estimation and high resolution, respectively), while simultaneously compensating for their drawbacks (i.e., poor resolution and texture, and sensitiveness to poorly illuminated scenarios, respectively). It is worth clarifying that the LIDAR and camera data are not fused at the image level. Instead, partial pose information is obtained by independently processing the acquired 3D and 2D datasets.

A Photonic Mixer Device (PMD), i.e., a 3D TOF camera, collects range images which are processed to measure the target distance along the sensor boresight direction and the cross-boresight relative rotations (pitch and yaw). The remaining 3 DOFs, i.e., the cross-boresight components of the relative position vector and the relative rotation around the boresight axis (roll), are obtained by processing monocular images from a high-resolution grayscale camera. The proposed algorithm compares the acquired range and monocular images to a target model represented as a set of plane and outer lines. Hence, the camera and the PMD perform edge-tracking and plane-tracking, respectively. This technique is extensively tested within the DLR facility called EPOS (European Proximity Operations Simulator) using a scaled mock-up of the rear part of a Geostationary Earth Orbit (GEO) satellite as target. The algorithm shows the capability of operating in real time at about 60 Hz. Millimeter- and

Table 8
Survey of 3D pose determination algorithms dealing with non-cooperative space targets.

Pub. year(s) Reference(s)	Pose acquisition technique	Pose tracking technique	Performance analysis criterion
2005 [147]	3D Binary TM	ICP algorithm	Off-line runs on synthetic and real point clouds
2008 [130]	Polygonal Aspect Hashing	ICP algorithm	Off-line runs on synthetic and real point clouds
2013 [156]	TM	ICP algorithm	Off-line runs on synthetic point clouds
2015–2017 [104,131,149]	3D On-line TM [104,131]	ICP algorithm	Off-line runs on synthetic point clouds
2016 [153]	3D On-line TM + PCA [149]		
2016 [150]	Branch-and-Bound algorithm [153] OUR-CVFH descriptor [150]	ICP algorithm ICP algorithm	Off-line runs on real images from ground-based experiments Off-line runs on synthetic point clouds

centimeter-level accuracy (in terms of mean and standard deviation) are achieved in the estimation of the cross-boresight and boresight components of the relative position vector, respectively. Instead the rotational parameters are characterized by a roll error (i.e., higher than 1° in terms of mean and standard deviation) slightly larger than the one obtained for pitch and yaw.

4.1.4. Summary

A review of the state of the art of monocular and 3D approaches for pose determination of an uncooperative space target has been presented. These techniques rely on the generation, either off-line or directly on-board, of a target model which provides information about shape and size of the observed space object. A critical design aspect to consider is the level of detail to be introduced in the target model. Specifically, it should be selected as a trade-off between the necessity to have information suitable for solving pose ambiguities, on one side, and to limit computational and storage on-board resources, on the other side. Despite promising results, demonstrated through numerical analyses and experimental test campaigns (ground- and/or space-based), some open challenges can be highlighted regarding the entire pose determination process.

Pose acquisition has proved to be the most critical task due to the necessity to search for an initial guess in the entire 6-DOF pose space. Hence, a list of the related unresolved issues is provided below.

- 1) Necessity to limit the computational load. This is particularly critical when dealing with fast target-chaser relative dynamics.
- 2) Necessity to enhance robustness against variability of pose conditions. This task is particularly critical since space objects typically have symmetric or nearly-symmetric shapes, thus being prone to produce ambiguous measurements.

Regarding pose tracking, edge-tracking and ICP-based techniques, used on images and 3D data, respectively, have demonstrated to be reliable, accurate (centimetric and sub-degree accuracy in relative position and attitude, respectively) and fast, provided that the initial guess from pose acquisition does not fall outside their specific field of convergence.

5. Uncooperative pose determination: unknown targets

Due to the peculiarity of the problem, the attention is focused directly to space applications. When dealing with unknown objects in space, e.g., debris or asteroids, one basic solution is to build the model directly on board using the available measurements. This could be done either during a prior monitoring stage (at a safe distance from the target), thus being able to apply reliable model-based algorithms for more critical maneuvers (close range rendezvous, final approach and docking), or simultaneously with pose estimation. This latter task can be accomplished by purposely adapting the well-known concept of Simultaneous Localization and Mapping (SLAM) [161], inherited from the robotic community. SLAM is the real-time process by which a mobile ground/aerial robot, moving within an unknown environment, computes its own trajectory (localization) while simultaneously building a map of that environment (mapping). In most applications, the environment is assumed to be static (static-world assumptions). However, observability issues can still be tackled if the non-static scene is composed of rigid objects (while the problem of performing SLAM within deformable large-scale environments is largely unexplored [162]). This latter assumption can be adopted for space applications involving an autonomous spacecraft flying in close-proximity of an unknown target (often called Resident Space Object, RSO). Indeed, the RSO has a specific orbital and rotational motion with respect to the relative navigation sensor, but it can be treated as a rigid object. In this context, SLAM algorithms relying on either monocular or 3D sensors are designed and tested.

The first approach to estimate simultaneously dynamic state,

geometric shape, and mass model parameters of an unknown target is proposed by Lichter and Dubowsky [163]. Specifically, this method relies on a set of range images acquired by a set of cooperating 3D sensors (either passive or active) which are assumed to be distributed uniformly around the target. The estimation architecture is composed of three blocks. First, a coarse estimate of the pose parameters is obtained. Specifically, the relative position is estimated by exploiting either the centroiding approach (i.e., the position vector is computed as the centroid of the acquired range image) or the concept of voxel occupancy (i.e., the space is discretized into voxels, which are volume elements whose occupancy level is assigned as a function of the number of enclosed range measurements). Instead, a rough initial estimate of the relative attitude is obtained by estimating the orientation of the principal axes of point- or voxel-based representation of the target. This pose information is used to feed a Kalman filter block which implements two separate filters for the estimation of the relative rotational and translational state. It is worth outlining that the rotational state vector includes the target principal inertia moments for shape estimation purposes. Both the rotational and translational process equations are written assuming the target to be a free-floating rigid object (i.e., no external forces are considered). Finally, a target model is generated using a standard stochastic map-building approach, for which well-assessed solutions exist in the literature [164]. This method can also be used to estimate the model frequencies of a flexible oscillating space structures as shown in Ref. [165]. Performance assessment is carried out with numerical simulations. This method is suitable for fast real-time implementation since it does not rely on feature detection and optical flow techniques which may be highly prone to errors and mismatching due to occlusions, harsh lighting conditions and presence of strongly reflective materials. However, it shall be considered that it relies on some stringent assumptions. Specifically, the adopted sensors are synchronized (i.e., no latencies are considered in the data exchange process) and their relative states are known without relative navigation uncertainties.

A feature-based SLAM algorithm relying on an EKF filter is proposed by Sonnenburg et al. [166]. As this work focuses exclusively on the filtering part of the relative navigation architecture, a set of 3D features, detected and tracked by a stereovision system, is assumed to be given as an input to the EKF. Performance of this approach is assessed by means of numerical simulations. Differently from Lichter's work, the process equation in the translational part of the KF is written by exploiting the well-known Hill's relative motion model, which assumes a circular target/chaser relative orbit. However, the filter state does not include information about the target inertia matrix. Two different case studies are considered for the target absolute rotational dynamics. In both the cases, the difference between the orientation of the target and chaser Local Vertical Local Horizontal (LVLH) reference frames is supposed negligible. Reference [166] also suggests the possibility of using the proposed EKF architecture considering features detected and tracked by a monocular camera. Indeed, the problem of the unobservability of the features' scale can be dealt with by using the inverse depth parameterization method [167].

A purely monocular based vision SLAM algorithm for pose tracking and shape reconstruction of an unknown target is presented by Augenstein and Rock [168]. Specifically, Bayesian filtering and measurement inversion approaches are combined for the relative rotational and translation dynamics, respectively, within a Rao-Blackwellized particle filter architecture. Bayesian filtering is used for the relative rotational dynamics but using a very simple model for the process equation obtained assuming constant rotational velocity (the time derivative of the angular momentum vector in the rotating body's frame in Euler's equation is set equal to zero). Again, no information about the target inertia moments is estimated since the target map is generated as a set of discrete 3D points describing its surface. Instead, the measurement inversion method is used for the translational portion of the relative navigation filter. In this case, the initial pose is assumed to be known in order to provide the filter with a prior estimate of the target map. Performance

assessment is done by means of both numerical simulation, as well as experimental tests considering an underwater tethered target.

Schnitzer et al. [169] combine the EKF-based SLAM filter described in Ref. [166] with a RANSAC-based surface reconstruction algorithm to generate a model of the unknown target which can be used for a further refinement of the estimated pose. Stereo images are used to estimate the 3D position of landmarks on the target surfaces. These landmarks correspond to natural features tracked along a sequence of frames exploiting the SURF descriptor [170]. Laboratory experiments are carried out with the Mini Proximity Operation Simulator testbed to evaluate performance of this approach. The surface reconstruction algorithm provides valuable results despite dealing with noisy, sparse and non-uniformly distributed point clouds. However, computational time is of a few seconds order, which means that real-time applicability of the shape reconstruction technique shall be further analyzed.

Tweddle, in his PhD dissertation [171], proposes a solution to solve the SLAM problem for an unknown space target spinning around any of its principal inertia axes by using a smoothing approach which relies on the work by Kaess et al. [172]. This choice is driven by the fact that smoothing ensures convergence capability better than standard filtering techniques, despite a heavier computational burden. Of course, this latter aspect may be an issue for real-time autonomous operations in orbit. The algorithm exploits 3D measurements from a stereo camera and allows estimating the entire relative state, including linear and angular velocities of the object as well as its center of mass, principal axes and diagonal inertia matrix. Indeed, the possibility of determining the principal inertia moments up to a scale factor (for a rigid body in absence of external torques) is demonstrated by means of a proper observability analysis. The applicability of the approach is verified by means of experimental tests conducted integrating a stereo system (Goggles) with the SPHERES satellites [96]. It is important to highlight the main assumptions made by the authors.

- 1) The chaser is assumed to be stationary.
- 2) The process model equations are obtained without including either a relative motion model or external torques for the relative translational and rotational dynamics, respectively

Benninghoff and Boge [173] describe an original method for the estimation of the inertia matrix and the center of mass of an unknown target. Again, the inertia moments are estimated up to a scale factor since the target is assumed to be a rigid body not subject to any external torque. Main limitation of this method is that it requires in input the relative position of the target geometric center (which would be coincident to the center of mass for a homogeneous target) and the relative attitude of the TRF with respect to an Inertial Reference Frame. These quantities are assumed to be determined using either a passive camera or an active LIDAR. Also, the target translational and angular velocities are assumed as determined by numerical differentiation from the position and attitude information. Numerical simulations demonstrate that the proposed approach is able to ensure satisfying accuracy level, provided that the target observation time is large enough. On this basis, the necessity of an inspection maneuver around the target prior to final approach and docking has been suggested.

A SLAM approach based on multiple iterated extended Kalman filters (IEKFs) and a maximum a posteriori (MAP) inertia tensor identification algorithm is proposed by Segal et al. [174]. The chaser spacecraft is assumed to be equipped with a stereovision system capable of providing estimates of the 3D position in CRF of a certain number of natural features located on the target surface. An interesting contribution is given by the fact that the process equation in the filtering architectures exploits a relative motion model based on the Hill's equations, which takes the kinematic coupling between translational and rotational dynamics into account [175]. Since the presence of external torques is considered in the rotational dynamics model, the inertia matrix can be estimated without any scale-factor ambiguity. The filtering architecture consists in running

in parallel multiple IEKFs considering a fixed number (e.g., 5) of hypothesis for the inertia parameters. This aspect poses some issues regarding the real-time applicability of this approach. Performance assessment is carried out by means of numerical simulations. The following assumptions are introduced regarding the achievable performance of the considered stereovision system. First, a fixed number of 10 features, uniformly distributed on the target surface, are assumed as perfectly tracked through a sequence of stereo-images. Second, a target/chaser relative distance larger than 100 m is considered in the simulated relative motion even though the baseline of the stereo configuration is short (i.e., 1 m) and the level of Gaussian noise added to the measurements is low (i.e., 10^{-5} rad). Additional results are obtained by an experimental laboratory campaign. For instance, the SLAM algorithm provides, on average, centimeter-level accuracy in the relative position with the sensor (i.e., a stereo camera with 14.5-cm-baseline) located at few meters. Of course, the scalability of these results for on-orbit implementation shall be carefully verified.

A navigation strategy for spacecraft operations in proximity of a small body is presented by Conway et al. [176]. One relevant original aspect is given by the fact that the implemented feature-based SLAM algorithm relies on measurements provided by a RGB-depth camera, which can acquire high-rate (up to 30 Hz) sequences of co-registered depth and color images. On this basis, the 3D model of the observed target is built as a set of landmarks, whose 3D position in the sensor reference frame is obtained by adding range information extracted from the depth map to the corresponding natural 2D features tracked over a long-enough sequence of color frames. Specifically, the ORB descriptor is used for feature detection and matching [177]. Like for several SLAM approaches applied to flying or ground robots, the map is updated at lower data rate with respect to the computation of the relative state (key-frame strategy [178]). Both numerical and experimental tests are carried out for preliminary performance assessment of this method, using the Microsoft Kinect as RGB-depth sensor.

A customized IEKF method for relative motion estimation and inertia parameters reconstruction of an unknown RSO is proposed by Pesce et al. [179]. The estimation architecture of the relative state is similar to [171], but the use of the IEKF allows improving the computational efficiency with respect to the smoothing approach. Due to the absence of external torques in the adopted relative rotational dynamics model, only the inertia ratios are estimated. However, a complete reconstruction of the inertia properties can be done by means of an off-line image processing method. The performance achieved by the SLAM approach is evaluated within a simulation environment adopting simplifying assumptions on the results of the stereo measurement process, similarly to [174]. Also, some experimental tests are carried out to preliminarily assess performance of the off-line method to reconstruct the inertia properties.

A summary of the main reviewed SLAM approaches for pose determination of uncooperative, unknown space objects is provided in Table 9.

It can be stated that the problem of estimating the relative state while simultaneously building a model of an unknown RSO using spaceborne processing resources still presents open issues. For instance, despite the availability of several filtering/smoothing architectures providing various levels of performance, the capabilities and limitation of the distinct types of EO sensors suitable for relative navigation are not always considered in a consistent way. Another interesting aspect to underline is given by the possibility to include more realistic relative motion models for both the rotational and translational dynamics in the process equation of the selected filtering/smoothing architecture.

6. Conclusion

A detailed review of the state of the art regarding techniques and algorithms enabling an active spacecraft (chaser) to carry out pose determination of a target space object is presented. This task is relevant to mission scenarios, i.e., formation flying, on-orbit servicing, active

Table 9

Survey of SLAM approaches for pose determination dealing with unknown, non-cooperative space targets.

Pub. year(s) Reference(s)	Sensor approach	SLAM approach	Performance analysis criterion
2004 [163]	Multiple cooperating LIDARs or stereo cameras	Two separate Kalman Filters for state estimation [163] Stochastic map-building [164]	Numerical simulations
2010 [166]	Stereo cameras	Feature-based EKF [166]	Numerical simulations
2011 [168]	Monocular camera	Rao-Blackwellized particle filter [168] Bayesian filter Partial measurement inversion	Numerical simulations Ground-based experimental tests
2012 [169]	Stereo cameras	Feature-based EKF [166] RANSAC-based surface reconstruction [169]	Ground-based experimental tests
2013 [171]	Stereo cameras	Smoothing algorithm [172]	Experimental tests on board of the ISS
2014 [174]	Stereo cameras	Multiple IEKF [174] MAP inertia tensor identification [174]	Numerical simulations Ground-based experimental tests
2014 [176]	RGB-depth camera	ORB-based EKF [176]	Numerical simulations Ground-based experimental tests
2017 [179]	Stereo cameras	IEKF [179]	Numerical simulations

debris removal and comet/asteroid exploration, which involve the chaser and the target to orbit in close-proximity.

First, the main technological solutions suitable for pose determination and relative navigation are classified. Radio-frequency, satellite-navigation, and electro-optical sensors may all represent valid options depending on the level of cooperativeness of the target, imposed by the considered mission scenario. Among the above options, electro-optical sensors surely represent the best one in terms of operational flexibility, since they can be used to estimate the target/chaser pose parameters in close-proximity operations, regardless of target cooperativeness level.

The main advantages and drawbacks related to the use of active or passive electro-optical sensors for pose determination tasks are also highlighted. This information is useful to carry out trade-off studies in terms of spatial resolution, update rate, robustness to visibility conditions, and on-board resources for mass, electrical and processing power. To summarize, LIDAR are more robust than passive sensors when dealing with poorly illuminated scenarios, even though they are typically heavier, more expensive and power consuming. Also, they provide 3D information about the target not achievable with monocular sensors, ensuring at the same time an operational range larger than stereo cameras. On the other hand, monocular and stereo cameras can provide finer resolution and larger update rate with respect to LIDAR. The limited LIDAR resolution can be partially compensated if LIDAR intensity data are processed together with the positional data to enrich the amount of information contained in the measured datasets. Recently, the possibility to combine the complementary advantages of LIDAR and passive sensors has emerged, and it is under study by researchers. The main technical specifications of monocular sensors, stereo cameras and LIDARs suitable for spacecraft pose determination are presented.

The pose determination problem is then analyzed considering the case of cooperative targets, cooperative, known targets and cooperative, unknown targets. For each of these cases, the main related methodological and theoretical issues are explained and a detailed overview of space-related implementations is discussed.

For monocular and 3D sensors, cooperative pose determination critically relies on the design of easily recognizable markers, arranged on the target surface according to unique patterns. Specifically, both the individual markers and the pattern need to be highly invariant to the

variations of illumination and pose conditions. This means that if an outlier-free solution to the correspondence problem can be found, several existing techniques can be used to extract the pose parameters. The choice of markers and pose determination approaches affects the achievable accuracy level and operational range.

Several monocular and 3D techniques are presented for pose determination of uncooperative, known targets, whose performance is assessed by means of numerical simulations and/or experimental test campaigns (on ground and/or in orbit). Although 3D techniques demonstrate to be a more mature solution, some common open issues are identified. Among them, the most critical ones are the large computational time needed to find an initial guess for the pose parameters, and the limited robustness of both the acquisition and tracking steps against variability of pose conditions. In view of these challenges, future research activities should be devoted to developing and testing techniques for autonomous detection of failures of both acquisition and tracking algorithms. In this respect, although preliminary concepts can be found in the literature, further investigations should be envisaged. Another interesting solution to deal with the sensitivity of pose determination performance against unfavorable observability conditions is the design of specific relative trajectories for close-proximity maneuvers which ensure target optimal viewing conditions for pose determination.

Finally, the problem of pose determination of uncooperative, unknown targets can be addressed by purposely adapting the concept of simultaneous localization and mapping, inherited by the robotic community, to space applications. This means that pose determination and target model generation are carried out at the same time by exploiting filtering (e.g., extended Kalman filters and particle filters) or smoothing architectures. The main solutions proposed in the open literature are discussed. This analysis shows that strong efforts should be devoted to including realistic models for the target/chaser rotational and translational relative dynamics within the developed filtering and smoothing architecture.

References

- [1] K. Alfriend, S.R. Vadali, P. Gurfil, J. How, L. Breger, *Spacecraft Formation Flying: Dynamics, Control and Navigation*, vol. 2, Butterworth-Heinemann, 2009.
- [2] F. Bauer, J. Bristow, J. Carpenter, J. Garrison, K. Hartman, T. Lee, A. Long, D. Kelbel, V. Lu, J. How, F. Busse, P. Axelrad, M. Moreau, *Enabling spacecraft formation flying in any earth orbit through spaceborne GPS and enhanced autonomy technologies*, *Space Technol.* 20 (4) (2001) 175–185.
- [3] A. Tatch, N. Fitz-Coy, S. Gladun, *On-orbit servicing: a brief survey*, in: *Proceedings of the 2006 Performance Metrics for Intelligent Systems Workshop*, Gaithersburg, MD, USA, Aug. 2006, pp. 21–23.
- [4] A. Flores-Abad, O. Ma, K. Pham, S. Ulrich, *A review of space robotics technologies for on-orbit servicing*, *Prog. Astronaut. Sci.* 68 (2014) 1–26.
- [5] X. Clerc, I. Retat, *Astrium vision on space debris removal*, in: *Proceedings of the 63rd International Astronautical Congress*, Naples, Italy, 1–5 Oct. 2012.
- [6] C. Bonnal, J.M. Ruault, M.C. Desjean, *Active debris removal: recent progress and current trends*, *Acta Astronaut.* 85 (2013) 51–60.
- [7] T. Kubota, S. Sawai, T. Hashimoto, J.I. Kawaguchi, *Robotics and autonomous technology for asteroid sample return mission*, in: *Proceedings 12th International Conference on Advanced Robotics*, July 2005.
- [8] A.F. Cheng, *Near Earth Asteroid Rendezvous: Mission Summary*, *Asteroids III*, vol. 1, 2002, pp. 351–366.
- [9] W. Fehse, *The drivers for the approach strategy*, in: *Automated Rendezvous and Docking of Spacecraft*, Cambridge University Press, Cambridge, UK, 2003, pp. 124–126.
- [10] T. Weismüller, M. Lein, *GN&C technology demonstrated by the orbital express autonomous Rendezvous and capture sensor system*, in: *Proceedings of the 29th Annual American Astronautical Society Guidance and Control Conference*, Feb. 2006.
- [11] H. Benninghoff, T. Tzschichholz, T. Boge, G. Gaias, *Far range image processing method for autonomous tracking of an uncooperative target*, in: *Proceedings of the 12th Symposium on Advanced Space Technologies in Robotics and Automation*, Noordwijk, The Netherlands, May 2013.
- [12] P. Bodin, R. Noteborn, R. Larsson, T. Karlsson, S. D'Amico, J.S. Ardaens, M. Delpach, J.C. Berge, *PRISMA formation flying demonstrator: overview and conclusions from the nominal mission*, *Adv. Astronaut. Sci.* 144 (2012) 441–460.
- [13] J.S. Llorente, A. Agenjo, C. Carrascosa, C. de Negueruela, A. Mestreau-Garreau, A. Cropp, A. Santovincenzo, *PROBA-3: precise formation flying demonstration mission*, *Acta Astronaut.* 82 (1) (2013) 38–46.
- [14] W.A. Bamford, J.W. Mitchell, M.C. Southward, P.J. Baldwin, L.B. Winternitz, G.W. Heckler, R. Kurichh, S.L. Sirotzky, *GPS navigation for the magnetospheric*

- multi-scale mission, in: Proceedings of the 22nd International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS 2009), Savannah, GA, 2009, pp. 1447–1457.
- [15] D.C. Woffinden, D.K. Geller, Navigating the road to autonomous orbital rendezvous, *J. Spacecr. Rockets* 44 (4) (2007) 898–909.
 - [16] E.M. Hinman, D.M. Bushman, Soviet Automated Rendezvous and Docking System Overview, Presented at a Automated Rendezvous and Capture Review, 1991. Executive Summary.
 - [17] A. Renga, M. Grassi, U. Tancredi, Relative navigation in LEO by carrier-phase differential GPS with intersatellite ranging augmentation, *Int. J. Aerosp. Eng.* 2013 (2013), 627509, 11 pages.
 - [18] S. D'Amico, J.S. Ardaens, S. De Florio, Autonomous formation flying based on GPS—PRISMA flight results, *Acta Astronaut.* 82 (1) (2013) 69–79.
 - [19] D.C. Woffinden, D.K. Geller, Relative angles-only navigation and pose estimation for autonomous orbital rendezvous, *J. Guid. Control Dyn.* 30 (5) (2007) 1455–1469.
 - [20] <https://www.nasa.gov/press-release/nasa-awards-contract-for-refueling-mission-spacecraft>.
 - [21] B.L. Benedict, Investing in satellite life extension—fleet planning options for spacecraft owner/operators, in: AIAA SPACE 2014 Conference and Exposition, AIAA Paper 2014-4445, Aug. 2014.
 - [22] G.M. Koretsky, M.S. Taylor, J.F. Nicoll, A Tutorial on ElectroOptical/Infrared (EO/IR) Theory and Systems, Institute for Defense Analyses, January 2013, pp. 15–19. https://www.ida.org/idamedia/Corporate/Files/Publications/IDA_Documents/SED/ida-document-d-4642.pdf.
 - [23] L. Walker, Automated proximity operations using image-based relative navigation, in: Proceedings of the 26th Annual USU/AIAA Conference on Small Satellites, SSC12-VII-3, Logan, Utah, August 2012.
 - [24] M. Okutomi, T. Kanade, A multiple-baseline stereo, *IEEE Trans. Pattern Anal. Mach. Intell.* 15 (4) (1993) 353–363.
 - [25] T. Kasai, M. Oda, T. Suzuki, Results of the ETS-7 mission – rendezvous docking and space robotics experiments, in: Proceedings of the 5th International Symposium on Artificial Intelligence, Robotics and Automation in Space, ESTEC, Noordwijk (The Netherlands), ESA SP-440, June 1999, pp. 299–306.
 - [26] M. Mokuno, I. Kawano, T. Suzuki, In-Orbit Demonstration of Rendezvous Laser Radar for unmanned autonomous rendezvous docking, *IEEE Trans. Aerosp. Electron. Syst.* 40 (2) (2004) 617–626.
 - [27] M. Mokuno, I. Kawano, In-orbit demonstration of an optical navigation system for autonomous rendezvous docking, *AIAA J. Spacecr. Rockets* 48 (6) (2011) 1046–1054.
 - [28] T.E. Rumford, Demonstration of autonomous rendezvous technology (DART) project summary, in: Proceedings of SPIE Space Systems and Technology Operations vol. 5088, August 2003.
 - [29] R.B. Friend, Orbital express program summary and mission overview, in: Proceedings of SPIE Def. Security Symposium, April 2008, 695803–695803.
 - [30] S. Croomes, Overview of the Dart Mishap Investigation Results, 2006, pp. 1–10. NASA Report.
 - [31] A.F. Heaton, R.T. Howard, R.M. Pinson, Orbital express AVGS validation and calibration for automated rendezvous, in: AIAA Guidance, Navigation and Control Conference and Exhibit, Honolulu (USA), August 2008.
 - [32] R.T. Howard, A.F. Heaton, R.M. Pinson, C.L. Carrington, J.E. Lee, T.C. Bryan, B.A. Robertson, S.H. Spencer, J.E. Johnson, The advanced Video guidance sensor: orbital express and the next generation, in: AIP Conference Proceedings, vol. 969, January 2008, pp. 717–724.
 - [33] S. Persson, P. Bodin, E. Gill, J. Harr, J. Jorgensen, PRISMA – an autonomous formation flying mission, in: ESA Small Satellite Systems and Services Symposium, Sardinia (Italy), September 2006.
 - [34] H. Benninghoff, T. Tzschichholz, T. Boge, G. Gaias, Far range image processing method for autonomous tracking of an uncooperative target, in: Proceedings of the 12th Symposium on Advanced Space Technologies in Robotics and Automation, Noordwijk, The Netherlands, May, 2012.
 - [35] M. Benn, J.L. Jørgensen, Short range pose and position determination of spacecraft using a μ -advanced stellar compass, in: Proceedings of the 3rd International Symposium on Formation Flying, Missions and Technologies, Noordwijk, The Netherlands, Apr. 2008.
 - [36] P. Bodin, R. Noteborn, R. Larsson, T. Karlsson, S. D'Amico, J.S. Ardaens, M. Delpech, J.C. Berges, The prisma formation flying demonstrator: overview and conclusions from the nominal mission, *Adv. Astronaut. Sci.* 144 (2012) 441–460.
 - [37] M. Delpech, J.C. Berges, T. Karlsson, F. Malbet, Results of PRISMA/FFIORD extended mission and applicability to future formation flying and active debris removal missions, *Int. J. Space Sci. Eng.* 5 (1.4) (2013) 382–409.
 - [38] R.T. Howard, T.C. Bryan, J. Lee, B. Robertson, Next Generation Advanced Video Guidance Sensor Development and Test, Univelt, Inc, 2009. P. O. Box 28130 San Diego CA 92198–0130 USA.
 - [39] J.L. Jørgensen, C.C. Liebe, The advanced stellar compass, development and operations, *Acta Astronaut.* 39 (9) (1996) 775–783.
 - [40] P.S. Jørgensen, J.L. Jørgensen, T. Denver, P. van den Braembuche, The micro advanced stellar compass for ESA'S PROBA 2 mission, in: Proceedings of the 5th International Symposium of the IAA on Small Satellites for Earth Observation, 2005, pp. 299–303.
 - [41] G. Fasano, D. Accardo, M. Grassi, A Stereo-vision Based System for Autonomous Navigation of an In-orbit Servicing Platform, 2009. ISBN-10: 1-56347-971-0, AIAA Infotech@Aero-space 2009, Seattle, USA.
 - [42] S. D'Amico, M. Benn, J.L. Jørgensen, Pose estimation of an uncooperative spacecraft from actual space imagery, *Int. J. Space Sci. Eng.* 5 (2) (2014) 171–189.
 - [43] B.J. Naasz, R.D. Burns, S.Z. Queen, J. Van Eepoel, J. Hannah, E. Skelton, The HST SM4 relative navigation sensor system: overview and preliminary testing results from the flight robotics lab, *J. Astronaut. Sci.* 57 (1–2) (2009) 457–483.
 - [44] B.J. Naasz, J. Van Eepoel, S. Queen, C. Southward, J. Hannah, Flight results of the HST SM4 relative navigation sensor system, in: Proceedings in Advances in the Astronautical Sciences, vol. 137, 2010, pp. 723–744.
 - [45] J.M. Galante, J.V. Eepoel, M. Strube, N. Gill, M. Gonzalez, A. Hyslop, B. Patrick, Pose measurement performance of the argon relative navigation sensor suite in simulated flight conditions, in: AIAA Guidance Navigation and Control Conference, August 2012.
 - [46] C. Liu, W. Hu, Relative pose estimation for cylinder-shaped spacecrafts using single image, *IEEE Trans. Aerosp. Electron. Syst.* 50 (4) (2014) 3036–3056.
 - [47] X. Du, B. Liang, W. Xu, Y. Qiu, Pose measurement of large non-cooperative satellite based on collaborative cameras, *Acta Astronaut.* 68 (11) (2011) 2047–2065.
 - [48] F. Yu, Z. He, B. Qiao, X. Vu, Stereo-vision-based relative pose estimation for the rendezvous and docking of noncooperative satellites, *Math. Probl. Eng.* 461283 (2014) 12, <http://dx.doi.org/10.1155/2014/461283>.
 - [49] F. Blais, Review of 20 Years of range sensor development, *J. Electron. Imaging* 13 (2004) 231–240.
 - [50] S. Crosby, S.H. Kang, Object identification in 3D flash lidar images, *J. Pattern Recognit. Res.* 6 (2) (2011) 193–200.
 - [51] A. Kirmani, A. Colaco, F. Wong, V. Goyal, Exploiting sparsity in time-of-flight range acquisition using a single time-resolved sensor, *Opt. Express* 19 (22) (2011) 21485–21507.
 - [52] A. Kovács, Scanning Strategies for Imaging Arrays. SPIE Astronomical Telescopes and Instrumentation, International Society for Optics and Photonics, 2008.
 - [53] R.M. Willett, F.M. Roummel, J.M. Nichols, Compressed sensing for practical optical imaging systems: a tutorial, *Opt. Eng.* 50 (7) (2011) 1–13.
 - [54] J.A. Christian, S. Cryan, A survey of LIDAR technology and its use in spacecraft relative navigation, in: AIAA Guidance, Navigation, and Control Conference, 7061, San Diego, CA, USA, Aug. 2008.
 - [55] C. English, S. Zhu, C. Smith, S. Ruel, I. Christie, Tridar: a hybrid sensor for exploiting the complementary nature of triangulation and LIDAR technologies, in: Proceedings of the 8th International Symposium on Artificial Intelligence, Robotics and Automation in Space, Sept. 2008.
 - [56] J.A. Beraldin, F. Blais, M. Rioux, L. Cournoyer, D. Laurin, S.G. MacLean, Eye-safe digital 3-D sensing for space applications, *Opt. Eng.* 39 (1) (2000) 196–211.
 - [57] J.A. Beraldin, F. Blais, M. Rioux, L. Cournoyer, D.G. Laurin, S.G. MacLean, Short and medium range 3D sensing for space applications, in: Proceedings of SPIE Visual Information Processing VI, vol. 3074, Jul. 1997, pp. 29–46.
 - [58] D.G. Laurin, J.A. Beraldin, F. Blais, M. Rioux, L. Cournoyer, A three dimensional tracking and imaging laser scanner for space operations, in: Proceedings of SPIE Laser Radar Technology and Applications IV, vol. 3707, May 1999, pp. 278–289.
 - [59] C.L. Smithpeter, R.O. Nellums, S.M. Lebien, G. Studor, G. James, LADAR measurements of the international space station, in: Proceedings of SPIE Laser Radar Technology and Applications VI, vol. 4377, Sept. 2001, pp. 65–72.
 - [60] S. Ruel, C. English, M. Anttil, P. Church, 3DLASSO: real-time pose estimation from 3D data for autonomous satellite servicing, in: Proceedings of the 2005 International Symposium on Artificial Intelligence for Robotics and Automation in Space, Munich, Germany, Sept. 2005.
 - [61] C. Samson, C. English, A. Deslauriers, I. Christie, F. Blais, F. Ferrie, Neptec 3D Laser Camera System: from space mission STS-105 to terrestrial applications, in: Proceedings of the 2002 ASTRO Conference, Ottawa, Canada, Nov. 2002.
 - [62] C.C. Liebe, A. Abramovici, R.K. Bartman, R.L. Bunker, J. Chapsky, C.C. Chu, D. Clouse, J.W. Dillon, B. Hausmann, H. Hemmati, R.P. Komfeld, C. Kwa, S. Mobasser, M. Newell, C. Padgett, W.T. Roberts, G. Spiers, Z. Warfield, M. Wright, Laser radar for spacecraft guidance applications, in: Proceedings of 2003 IEEE Aerospace Conference, vol. 6, Mar. 2003, pp. 2647–2662. Big Sky, MT, USA.
 - [63] R.P. Kornfeld, R.L. Bunker, G.C. Cucullu, J.C. Essmiller, F.Y. Hadaegh, C.C. Liebe, C.W. Padgett, E.C. Wong, New millennium ST6 autonomous rendezvous experiment (ARX), in: Proceedings of the 2003 IEEE Aerospace Conference, vol. 1, Mar. 2003, pp. 369–380. Big Sky, MT, USA.
 - [64] <http://www.jena-optronik.de/en/aocs/rvs.html>.
 - [65] <http://www.advancedscientificconcepts.com/products/overview.html>.
 - [66] L.K. Miller, J. Masciarelli, R. Rohrschneider, I. Gravseth, Critical advancement in telerobotic servicing vision technology, in: AIAA SPACE 2010 Conference & Exposition, 2010.
 - [67] D.S. Lauretta, OSIRIS-REx asteroid sample-return mission, in: Handbook of Cosmic Hazards and Planetary Defense, 2015, pp. 543–567.
 - [68] J.A. Christian, M. Patangan, H. Hinkel, K. Chevray, J. Brazzel, Comparison of orion vision navigation sensor performance from STD-134 and the space operations simulation center, in: AIAA Guidance, Navigation, and Control Conference, Minneapolis, MN, USA, Aug. 2012.
 - [69] R.Y. Tsai, A versatile camera calibration technique for high-accuracy 3D machine vision metrology using off-the-shelf TV cameras and lenses, *IEEE J. Robot. Autom.* 3 (4) (1987) 323–344.
 - [70] M.A. Fischler, R.C. Bolles, Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography, *Commun. ACM* 24 (6) (1981) 381–395.
 - [71] Y. Wu, Z. Hu, PnP problem revisited, *J. Math. Imaging Vis.* 24 (1) (2006) 131–141.
 - [72] X.S. Gao, X.R. Hou, J. Tang, H.F. Cheng, Complete solution classification for the perspective-three-point problem, *IEEE Trans. Pattern Anal. Mach. Intell.* 25 (8) (2003) 930–943.

- [73] M.A. Abidi, T. Chandra, A new efficient and direct solution for pose estimation using quadrangular targets: algorithm and evaluation, *IEEE Trans. Pattern Anal. Mach. Intell.* 17 (5) (1995) 534–538.
- [74] L. Quan, Z. Lan, Linear n-point camera pose determination, *IEEE Trans. Pattern Anal. Mach. Intell.* 21 (8) (1999) 774–780.
- [75] V. Lepetit, F. Moreno-Noguer, P. Fua, Eppn: an accurate $O(n)$ solution to the pnp problem, *Int. J. Comput. Vis.* 81 (2) (2009) 155–166.
- [76] J.A. Hesch, S.I. Roumeliotis, A direct least-squares (DLS) method for PnP, in: *Proceedings of the IEEE International Conference on Computer Vision, Barcelona, Spain, Nov. 2011*, pp. 383–390.
- [77] L. Ferraz, X. Binefa, F. Moreno-Noguer, Very fast solution to the PnP problem with algebraic outlier rejection, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Columbus, OH, USA, Jun. 2014*, pp. 501–508.
- [78] M.D. Shuster, S. Oh, Three-axis attitude determination from vector observations, *J. Guid. Control, Dyn.* 4 (1981) 70–77.
- [79] I.Y. Bar-Itzhack, R.R. Harman, Optimized TRIAD algorithm for attitude determination, *J. Guid. Control Dyn.* 20 (1) (1997) 208–211.
- [80] F.L. Markley, Attitude determination using two vector measurements, in: *Proceedings, Flight Mechanics Symposium, NASA Goddard Space Flight Center, Greenbelt, MD, May 1999*, pp. 39–52. NASA Conference Publication NASA/CP-19989-209235.
- [81] S. Tanygin, M.D. Shuster, The many triad algorithms, in: *Proceedings of the AAS/AIAA 17th Space Flight Mechanics Meeting, Sedona, Arizona, Jan. 2007*, pp. 81–99.
- [82] F.L. Markley, M. Mortari, Quaternion attitude estimation using vector measurements, *J. Astronaut. Sci.* 48 (2000) 359–380.
- [83] Y. Cheng, M.D. Shuster, Improvement to the implementation of the QUEST algorithm, *J. Guid. Control, Dyn.* 37 (1) (2013) 301–305.
- [84] K.S. Arun, T.S. Huang, S.D. Blostein, Least-squares fitting of two 3-D point sets, *IEEE Trans. Pattern Anal. Mach. Intell.* 5 (1987) 698–700.
- [85] B.K. Horn, H.M. Hilden, S. Negahdaripour, Closed-form solution of absolute orientation using orthonormal matrices, *JOSA A* 5 (7) (1988) 1127–1135.
- [86] S. Umeyama, Least-squares estimation of transformation parameters between two point patterns, *IEEE Trans. Pattern Anal. Mach. Intell.* 13 (4) (1991) 376–380.
- [87] P. Besl, N. McKay, A method for registration of 3-D shapes, *IEEE Trans. Pattern Anal. Mach. Intell.* 14 (2) (1992) 239–256.
- [88] Y. Chen, G. Medioni, Object modelling by registration of multiple range images, *Image Vis. Comput.* 10 (3) (1992) 145–155.
- [89] J.E. LeCroy, D.S. Hallmark, R.T. Howard, Effects of optical artifacts in a laser-based spacecraft navigation sensor, in: *Proceedings of the SPIE Defense and Security Symposium, Apr. 2007*, 655507–655507.
- [90] L. Mullins, A.F. Heaton, J. Lomas, Advanced Video Guidance Sensor Inverse Perspective Algorithm, MSFC Internal Memo TD45 (04–09), Dec. 2003.
- [91] J. LeCroy, D. Hallmark, P. Scott, R.T. Howard, Comparison of navigation solutions for autonomous spacecraft from multiple sensor systems, in: *Proceedings of the 2008 SPIE Defense and Security Symposium, Apr. 2008*.
- [92] C. Becker, R.T. Howard, J. Rakoczy, Smartphone Video guidance sensor for small satellites, in: *Proceedings of the 27th AIAA/USU Conference on Small Satellites*, 2013.
- [93] B.E. Tweddle, A. Saenz-Otero, Relative computer vision-based navigation for small inspection spacecraft, *J. Guid. Control, Dyn.* 38 (5) (2014) 969–978.
- [94] L.B. Gatrell, W.A. Hoff, C.W. Sklair, Robust image features: concentric contrasting circles and their image extraction, in: *Proc. of Cooperative Intelligent Robotics in Space. Washington, USA, SPIE*, 1991.
- [95] R.M. Haralick, H. Joo, C.N. Lee, X. Zhuang, V.G. Vaidya, M.B. Kim, Pose estimation from corresponding point data, *IEEE Trans. Syst. Man, Cybern.* 19 (6) (1989) 1426–1446.
- [96] S. Mohan, A. Saenz-Otero, S. Nolet, D.W. Miller, S. Sell, SPHERES flight operations testing and execution, *Acta Astronaut.* 65 (7) (2009) 1121–1132.
- [97] M. Pertile, M. Mazzucato, L. Bottaro, S. Chiodini, S. Debei, F. Lorenzini, Uncertainty evaluation of a vision system for pose measurement of a spacecraft with fiducial markers, in: *Proceedings of the 2nd IEEE International Workshop on Metrology for Aerospace, Benevento, Italy, Jun. 2015*, pp. 283–288.
- [98] L. Kneip, D. Scaramuzza, R. Siegwart, A novel parametrization of the perspective-three-point problem for a direct computation of absolute camera position and orientation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Colorado Springs, CO, USA, Jun. 2011*.
- [99] J.A. Christian, S.B. Robinson, C.N. Souza, J.P. Ruiz, Cooperative relative navigation of spacecraft using flash light detection and ranging sensors, *J. Guid. Control, Dyn.* 37 (2) (2014) 452–465.
- [100] D. Conte, P. Foggia, C. Sansone, M. Vento, Thirty years of graph matching in pattern recognition, *Int. J. Pattern Recognit. Artif. Intell.* 18 (3) (2004) 265–298.
- [101] S.B. Robinson, J.A. Christian, Pattern design for 3D point matching, *Navigation* 62 (3) (2015) 189–203.
- [102] J. Junkins, M. Majji, B. Macomber, J. Davis, J. Doebbler, R. Noster, Small body proximity sensing with a novel HD3-d LADAR system, *Adv. Astronaut. Sci.* 141 (2011) 341–354.
- [103] J.M. Kelsey, J. Byrne, M. Cosgrove, S. Seereeram, R.K. Mehra, Vision-based relative pose estimation for autonomous rendezvous and docking, in: *Proceedings of the 2006 IEEE Aerospace Conference, Big Sky, MT, USA, Mar. 2006*.
- [104] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, Uncooperative pose estimation with a LIDAR-based system, *Acta Astronaut.* 110 (2015) pp.287–297.
- [105] P. David, D.F. DeMenthon, R. Duraiswami, H. Samet, SoftPOSIT: simultaneous pose and correspondence determination, *Int. J. Comput. Vis.* 59 (3) (2004) 259–284.
- [106] S. Gold, A. Rangarajan, A graduated assignment algorithm for graph matching, *IEEE Trans. Pattern Anal. Mach. Intell.* 18 (4) (1996) 377–388.
- [107] S. Gold, A. Rangarajan, C.P. Lu, S. Pappu, E. Mjolsness, New algorithms for 2d and 3d point matching: pose estimation and correspondence, *Pattern Recognit.* 31 (8) (1998) 1019–1031.
- [108] D.F. DeMenthon, P. Davis, Model-based object pose in 25 lines of code, *Int. J. Comput. Vis.* 15 (1–2) (1995) 123–141.
- [109] P. David, D.F. DeMenthon, R. Duraiswami, H. Samet, Simultaneous pose and correspondence determination using line features, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Madison, WI, USA, Jun. 2003*, pp. 424–431.
- [110] D.G. Lowe, Object recognition from local scale-invariant features, in: *Proceedings of the 7th IEEE International Conference on Computer Vision, Kerkyra, Greece, Sept. 1999*, pp. 1150–1157.
- [111] C. Doignon, An introduction to model-based pose estimation and 3-d tracking techniques, in: R. Stolkin (Ed.), *Scene Reconstruction, Pose Estimation and Tracking*, first ed., InTech, 2007, pp. 359–382 ch. 20.
- [112] D.G. Lowe, Three dimensional object recognition from single two dimensional images, *Artif. Intell.* 31 (1985) 355–395.
- [113] R.C. Gonzalez, R.E. Woods, *Digital Image Processing*, second ed., Prentice Hall, Upper Saddle River, NJ, USA, 2002.
- [114] N.N. Dawoud, B.B. Samir, J. Janier, Fast template matching method based optimized sum of absolute difference algorithm for face localization, *Int. J. Comput. Appl.* 18 (2011) 30–34.
- [115] K. Brieche, U.D. Hanebeck, Template matching using fast normalized cross correlation, in: *Proceedings of SPIE*, 2001, p. 4387, <http://dx.doi.org/10.1117/12.421129>.
- [116] D.M. Gavril, V. Philomin, Real-time object detection using distance transforms, in: *Proceedings of the IEEE International Conference on Intelligent Vehicles, Stuttgart, Germany, Oct. 1998*, pp. 274–279.
- [117] C. Reinbacher, M. Ruther, H. Bischof, Pose estimation of known objects by efficient silhouette matching, in: *Proceedings of the IEEE 20th International Conference on Pattern Recognition, Istanbul, Turkey, Aug. 2010*, pp. 1080–1083.
- [118] T. Drummond, R. Cipolla, Real-time visual tracking of complex structures, *IEEE Trans. Pattern Anal. Mach. Intell.* 24 (7) (2002) 932–946.
- [119] A.I. Comport, E. Marchand, M. Pressigout, F. Chaumette, Real-time markerless tracking for augmented reality: the virtual visual servoing framework, *IEEE Trans. Vis. Comput. Graph.* 12 (4) (2006) 615–628.
- [120] C. Liu, W. Hu, Relative pose estimation for cylinder-shaped spacecrafts using single image, *IEEE Trans. Aerosp. Electron. Syst.* 50 (4) (2016) 3036–3056.
- [121] T.F. Cootes, G.J. Edwards, C.J. Taylor, Active appearance models, *IEEE Trans. Pattern Anal. Mach. Intell.* 23 (6) (2001) 681–685.
- [122] S. Wold, K. Esbensen, P. Geladi, Principal component analysis, *Chemom. Intell. Lab. Syst.* 2 (1) (1987) 37–52.
- [123] P. Mittrapiyanuruk, G.N. DeSouza, A.C. Kak, Calculating the 3d-pose of rigid-objects using active appearance models, in: *Proceedings of the IEEE International Conference on Robotics and Automation, New Orleans, LA, USA, Apr. 2004*, pp. 5147–5152.
- [124] M.J. Black, A.D. Jepson, Eigenttracking: robust matching and tracking of articulated objects using a view-based representation, *Int. J. Comput. Vis.* 26 (1) (1998) 63–84.
- [125] A.E. Johnson, M. Hebert, Using spin images for efficient object recognition in cluttered 3D scenes, *IEEE Trans. Pattern Anal. Mach. Intell.* 21 (5) (1999) 433–449.
- [126] A. Flint, A. Dick, A. Van Den Hengel, Thrift: local 3d structure recognition, in: *Proceedings of IEEE Conference Digital Image Computing: Techniques and Applications, Adelaide, Australia, Dec. 2007*, pp. 182–188.
- [127] H.T. Ho, D. Gibbins, Multi-scale feature extraction for 3D models using local surface curvature, in: *Proceedings of IEEE Conference Digital Image Computing: Techniques and Applications, Canberra, Australia, Dec. 2008*, pp. 16–23.
- [128] Y. Guo, M. Bennamoun, F. Soheli, M. Lu, J. Wan, 3D object recognition in cluttered scenes with local surface features: a survey, *IEEE Trans. Pattern Anal. Mach. Intell.* 36 (11) (2014) 2270–2287.
- [129] Y. Guo, M. Bennamoun, F. Soheli, M. Lu, J. Wan, N.M. Kwok, A comprehensive performance evaluation of 3D local feature descriptors, *Int. J. Comput. Vis.* 116 (1) (2016) 66–89.
- [130] S. Ruel, T. Luu, M. Anctil, S. Gagnon, Target localization from 3d data for on-orbit autonomous rendezvous and docking, in: *Proceedings of the 2008 IEEE Aerospace Conference, Big Sky, MT, USA, Mar. 2008*.
- [131] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, A model-based 3D template matching technique for pose acquisition of an uncooperative space object, *Sensors* 15 (3) (2015) 6360–6382.
- [132] M. Greenspan, P. Jasiobedzki, Pose determination of a free-flying satellite, in: *Proceedings of the 2002 International Conference on Imaging Science, Systems, and Technology, Las Vegas, NV, USA, Jun. 2002*.
- [133] J. Salvi, C. Matabosch, D. Fofi, J. Forest, A review of recent range image registration methods with accuracy evaluation, *Image Vis. Comput.* 25 (5) (2007) 578–596.
- [134] S. Rusinkiewicz, M. Levoy, Efficient variants of the ICP algorithm, in: *Proc. of the 3rd IEEE International Conference on 3-D Digital Imaging and Modeling, Quebec City, Canada, Jun. 2001*, pp. 145–152.
- [135] M. Greenspan, L. Shang, P. Jasiobedzki, Efficient tracking with the bounded hough transform, in: *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Washington, DC, USA, Jun. 2004*, 1-520-1-527.

- [136] R.O. Duda, P.E. Hart, Use of the Hough transformation to detect lines and curves in pictures, *Commun. ACM* 15 (1) (1972) 11–15.
- [137] K. Kanani, A. Petit, E. Marchand, T. Chabot, B. Gerber, Vision based navigation for debris removal missions, in: *Proceedings of the 63rd International Astronautical Congress*, Naples, Italy, Oct. 2012.
- [138] A. Petit, E. Marchand, K. Kanani, Vision-based detection and tracking for space navigation in a rendezvous context, in: *Proceedings of the International Symposium on Artificial Intelligence, Robotics and Automation in Space*, Turin, Italy, Sept. 2012.
- [139] S. Sharma, S. D'amico, Comparative assessment of techniques for initial pose estimation using monocular vision, *Acta Astronaut.* 123 (2016) 435–445.
- [140] J.G. Semple, G.T. Kneebone, *Algebraic Projective Geometry*, Clarendon Press, London, 1952, pp. 72–129 ch. IV and V.
- [141] A. Cropp, P. Palmer, C.I. Underwood, Pose estimation of target satellite for proximity operations, in: *Proceedings of the 14th Annual AIAA/USU Conference on Small Satellites*, Paper no. SSC00-III-7, Logan, UT, USA, Aug. 2000.
- [142] C. Lee, R.M. Haralick, Statistical estimation for exterior orientation from line-to-line correspondences, *Image Vis. Comput.* 14 (1996) 379–388.
- [143] H. Zhang, Z. Jiang, A. Elgammal, Vision-based pose estimation for cooperative space objects, *Acta Astronaut.* 91 (2013) 115–122.
- [144] P. Gasbarri, M. Sabatini, G.B. Palmerini, Ground tests for vision based determination and control of formation flying spacecraft trajectories, *Acta Astronaut.* 102 (2014) 378–391.
- [145] M.A. Rodríguez, D.E. Olmos, J.G. Fernández, IP-based pose estimation for space debris removal: implementation and results, in: *Proceedings of the 65th International Astronautical Congress*, Toronto, Canada, Sept.-Oct. 2014.
- [146] J. Cai, P. Huang, B. Zhang, D. Wang, A TSR visual servoing system based on a novel dynamic template matching method, *Sensors* 15 (12) (2015) 32152–32167.
- [147] P. Jasiobedzki, S. Se, T. Pan, M. Umasuthan, M. Greenspan, Autonomous satellite rendezvous and docking using LIDAR and model based vision, in: *Proceedings of SPIE Spaceborne Sensor II*, vol. 5798, Jun. 2005.
- [148] P. Jasiobedzki, M. Greenspan, G. Roth, H. Ng, N. Witcomb, Video-based system for satellite proximity operations, in: *Proceedings of the 7th ESA Workshop on Advanced Space Technologies for Robotics and Automation*, Noordwijk, The Netherlands, Nov. 2002.
- [149] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, Pose Estimation for Spacecraft Relative Navigation Using Model-based Algorithms, *IEEE Trans. Aerosp. Electron. Syst.*, DOI: 10.1109/TAES.2017.2650785.
- [150] J.O. Woods, J.A. Christian, LIDAR-based relative navigation with respect to non-cooperative objects, *Acta Astronaut.* 126 (2016) 298–311.
- [151] R.B. Rusu, N. Blodow, M. Beetz, Fast point feature histograms (FPFH) for 3D registration, in: *Proceedings of the 2009 IEEE International Conference on Robotics and Automation*, May 2009, pp. 3212–3217.
- [152] A. Rhodes, E. Kim, J.A. Christian, T. Evans, LIDAR-based relative navigation of non-cooperative objects using point Cloud Descriptors. *Proceedings of the 2016 AIAA/AAS Astrodynamics Specialist Conference*.
- [153] L. Liu, G. Zhao, Y. Bo, Point cloud based relative pose estimation of a satellite in close range, *Sensors* 16 (6) (2016) 824.
- [154] J. Yang, H. Li, Y. Jia, Go-icp: solving 3d registration efficiently and globally optimally. *Proceedings of the 2013 IEEE International Conference on Computer Vision*, pp. 1457–1464.
- [155] J. Ventura, A. Fleischner, U. Walter, Pose tracking of a noncooperative spacecraft during docking maneuvers using a time-of-flight sensor, in: *AIAA Guidance, Navigation and Control Conference*, Jan. 2016. AIAA Paper 2016-0875.
- [156] Sommer, I. Ahrens, GNC for a rendezvous in space with an uncooperative target, in: *5th International Conference on Spacecraft Formation Flying Missions and Technologies*, Germany, Munich, May 2013.
- [157] S.I. Nishida, S. Kawamoto, Y. Okawa, F. Terui, S. Kitamura, Space debris removal system using a small satellite, *Acta Astronaut.* 65 (1) (2009) 95–102.
- [158] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, Performance evaluation of 3D model-based techniques for autonomous pose initialization and tracking, in: *Proceedings of the AIAA SciTech 2015*, Kissimmee, USA, Jan. 2015.
- [159] L. Shang, P. Jasiobedzki, M. Greenspan, Model-based tracking by classification in a tiny discrete pose space, *IEEE Trans. Pattern Anal. Mach. Intell.* 29 (6) (2007) pp.976–989.
- [160] T. Tzschichholz, T. Boge, K. Schilling, Relative pose estimation of satellites using PMD-/CCD-sensor data fusion, *Acta Astronaut.* 109 (2015) 25–33.
- [161] H. Durrant-Whyte, T. Bailey, Simultaneous localization and mapping: part I, *IEEE Robot. Autom. Mag.* 13 (2) (2006) 99–110.
- [162] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, J.L. Leonard, Past, present, and future of simultaneous localization and mapping: toward the robust-perception age, *IEEE Trans. Robot.* 32 (6) (2016) 1309–1332.
- [163] M.D. Lichter, S. Dubowsky, State, shape, and parameter estimation of space objects from range images, in: *Proceedings of the 2004 IEEE International Conference on Robotics and Automation*, Apr. 2004.
- [164] R.C. Smith, P. Cheeseman, On the representation and estimation of spatial uncertainty, *Int. J. Robot. Res.* 5 (4) (1987) 56–68.
- [165] M.D. Lichter, S. Dubowsky, H. Ueno, S. Mitani, Shape, motion, and parameter estimation of flexible space structures using laser rangefinders, in: *Proceedings of Robotics: Science and Systems*, Jun. 2005.
- [166] A. Sonnenburg, M. Tkocz, K. Janschek, EKF-slam based approach for spacecraft rendezvous navigation with unknown target spacecraft, *IFAC Proc. Vol.* 43 (15) (2010) 339–344.
- [167] J. Civera, A.J. Davison, J.M. Montiel, Inverse depth parametrization for monocular SLAM, *IEEE Trans. Robot.* 24 (5) (2008) 932–945.
- [168] S. Augenstein, S.M. Rock, Improved frame-to-frame pose tracking during vision-only SLAM/SFM with a tumbling target, in: *Proceedings of the IEEE International Conference on Robotics and Automation*, May 2011.
- [169] F. Schnitzer, K. Janschek, G. Willich, Experimental results for image-based geometrical reconstruction for spacecraft Rendezvous navigation with unknown and uncooperative target spacecraft, in: *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, Oct. 2012.
- [170] H. Bay, A. Ess, T. Tuytelaars, L.V. Gool, Speeded-up robust features (SURF), *Comput. Vis. Image Underst.* 110 (2008) 346–359.
- [171] B.E. Tweddle, Computer Vision-based Localization and Mapping of an Unknown, Uncooperative and Spinning Target for Spacecraft Proximity Operations, PhD, Massachusetts Institute of Technology, Cambridge, MA, 2013.
- [172] M. Kaess, A. Ranganathan, F. Dellaert, ISAM: incremental smoothing and mapping, *IEEE Trans. Robot.* 24 (6) (2008) 1365–1378.
- [173] H. Benninghoff, T. Boge, Rendezvous involving a non-cooperative, tumbling target - estimation of moments of inertia and center of mass of an unknown target, in: *Proceedings of the 25th International Symposium on Space Flight Dynamics*, Munich, Germany, Oct. 2015.
- [174] S. Segal, A. Carmi, P. Gurfil, Stereovision-based estimation of relative dynamics between noncooperative satellites: theory and experiments, *IEEE Trans. Control Syst. Technol.* 22 (2) (2014) 568–584.
- [175] S. Segal, P. Gurfil, Effect of kinematic rotation-translation coupling on relative spacecraft translational dynamics, *J. Guid. Control Dyn.* 32 (3) (2009) 1045–1050.
- [176] D. Conway, B. Macomber, K.A. Cavalieri, J.L. Junkins, Vision-based relative navigation filter for asteroid rendezvous, in: *Proceedings of the AAS Guidance, Navigation, and Control Conference*, Breckenridge, CO, Feb. 2014.
- [177] E. Rublee, V. Rabaud, K. Konolige, G. Bradski, ORB: an efficient alternative to SIFT or SURF, in: *Proceedings of the IEEE International Conference Computer Vision*, Nov. 2011.
- [178] F. Andert, N. Ammann, J. Puschel, J. Dittrich, On the safe navigation problem for unmanned aircraft: visual odometry and alignment optimizations for UAV positioning, in: *Proceedings of the 2014 IEEE International Conference on Unmanned Aircraft Systems*, May 2014.
- [179] V. Pesce, M. Lavagna, R. Bevilacqua, Stereovision-based pose and inertia estimation of unknown and uncooperative space objects, *Adv. Space Res.* 59 (1) (2017) 236–251.